

ORIGINAL ARTICLE

ON DISTRIBUTIONAL AUTOREGRESSION AND ITERATED TRANSPORTATION

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We consider the problem of defining and fitting models of autoregressive time series of probability distributions on a compact interval of $\mathbb{R}$. An order-1 autoregressive model in this context is to be understood as a Markov chain, where one specifies a certain structure (regression) for the one-step conditional Fréchet mean with respect to a natural probability metric. We construct and explore different models based on iterated random function systems of optimal transport maps. While the properties and interpretation of these models depend on how they relate to the iterated transport system, they can all be analyzed theoretically in a unified way. We present such a theoretical analysis, including convergence rates, and illustrate our methodology using real and simulated data. Our approach generalizes or extends certain existing models of transportation-based regression and autoregression, and in doing so also provides some additional insights on existing models.

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1. INTRODUCTION

In distributional regression, one aims to describe/estimate the relationship between a response distribution ν , and a covariate distribution μ , viewed as random measures. This is to be done on the basis of an i.i.d. sample of random pairs $\{(\mu_i, \nu_i)\}_{i=1}^n$. The relationship is modeled *globally*, in that the complete distributions (seen as random elements of a suitable function space) are being related. In this sense, such models are useful in contexts where samples from several distributions are available, but these samples are not directly related – so we are sampling several univariate marginal distributions rather than a single multi-variate joint distribution. This can be due to data collection limitations, or simply because there is no natural joint law.

In light of this global perspective, distributional regression falls under the label of functional regression – where one random function is to be regressed on another (Morris, 2015). However, the nonlinear nature of probability distributions distinguishes distributional regression from typical functional regression. In usual function-on-function (auto)regression, one can model the regression via the usual (Bochner) conditional expectation and bounded linear transformations on Hilbert spaces (Imaizumi and Kato, 2018; Pham and Panaretos, 2018). But these concepts do not readily apply in distributional regression, where one is confronted with the challenges of geometrical data analysis (Patrangenaru and Ellingson, 2015; Petersen *et al.*, 2022). One approach to distributional regression circumvented this issue by embedding the distributions in a Hilbert space via suitable transformations (Kneip and Utikal, 2001; Delicado, 2011; Petersen and Müller, 2016; Kokoszka *et al.*, 2019). Another approach has focused on directly modeling random distributions as random elements of the Wasserstein space, a geodesic metric space

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related to optimal transport, increasingly seen as a canonical setting for distributional statistics (Panaretos and Zemel, 2020). In this context, Bochner integrals are replaced by Fréchet means (Panaretos and Zemel, 2016; Bigot *et al.*, 2018; Zemel and Panaretos, 2019), and what remains is the choice of regressor function, i.e., the specification of a relationship linking the conditional Fréchet mean of the response to the covariate.

Two general strategies have arisen for this specification. The geometrical approach uses the fact that the Wasserstein space is locally Hilbert-like, and defines classical Hilbertian regression by lifting covariate and response on an appropriate tangent space (see Chen *et al.* (2021) and Zhang *et al.* (2022)). While this model has a natural mathematical interpretation, its statistical interpretation is somewhat contrived. The other strategy is to directly specify the regression transformation as an optimal transport map, exploiting convexity and shape constraints, rather than geometrical features (see Ghodrati and Panaretos (2022)). This has the advantage of a clean interpretation and of avoiding ill-posedness issues.

Distributional autoregression is a natural next-step for distributional regression models – indeed, it is arguably the setting where most distributional regression data sets arise. Rather than i.i.d. covariate/response distributions, one observes a dependent sequence of probability distributions $\{\mu_n\}_{n=1}^N$. When viewed as a Markov chain in the Wasserstein space, this sequence can be modeled autoregressively by specifying a relationship between the conditional Fréchet mean at time $n + 1$ and the chain at time n . Once again, this can be done geometrically (as indeed was already explored in (Chen *et al.*, 2021) and (Zhang *et al.*, 2022)), or by way of optimal transport maps, with similar advantages/disadvantages.

A first contribution based directly on transport maps was made in Zhu and Müller (2021), where random perturbations of the identity were iteratively contracted/composed to form a time-dependent sequence. This was subsequently used either as ‘increments’ between consecutive distributions or as ‘deviations’ from the marginal Fréchet mean, to produce autoregressive models. Key in this approach was the use of iterated random function systems and a canny definition of a contraction operation on the space of transport maps, allowing to mimic the contractive effect of a correlation operator in usual autoregression. Jiang (2022) subsequently generalized this approach to autoregressive modeling to the case of vector-valued distributional chains, i.e., time-evolving vectors with distributions as coordinates.

A salient limitation of this approach is that the entire dynamics of the process reduce to a single scalar quantity $|\alpha| \leq 1$, regulating the ‘strength’ of the contraction. While this resembles real-valued autoregressive processes, it is likely too rigid in a functional context (or even a multi-variate context), and can have undesirable consequences when asserting stationarity (see Section 3.3 for a more extensive discussion). Ideally, a genuinely functional model would allow for a *functional* specification of the dynamics, thus capable of expressing more complex dependencies. In response to this drawback, Zhu and Müller (2021) also defined a model where the scalar contraction coefficient is replaced by a *functional contraction coefficient*, contracting variably across the domain. This comes with the caveat of a more complicated theory, including cumbersome technical assumptions, as well as a more involved interpretation.

The purpose of this article is to introduce and develop transportation-based autoregressive models with genuinely functional dynamics, yielding easily interpretable yet rich classes of distributional autoregressions (see Figure 1 for example realizations). To do so, we extend to the autoregressive case the functional structure of Ghodrati and Panaretos (2022), where the regression operator is a monotone rearrangement, making use of the scalar ‘contractive effect’ introduced by Zhu and Müller (2021) – intuitively, we posit a model where the *shape* of the dynamics is captured by a monotone map, modulated by a contractive parameter α regulating the degree of non-degeneracy of the model. In its simplest form, this approach can be interpreted as positing that

$$\mu_{n+1} = \theta_n \# [\alpha \mu_n], \quad n \in \mathbb{Z},$$

for i.i.d. random increasing maps θ_n with $\mathbb{E}[\theta_n(x)] = S(x)$; S a deterministic monotone map; $\#$ the pushforward operation, $f\#\mu(A) = \mu(f^{-1}(A))$; and $\mu_n \mapsto [\alpha \mu_n]$ a barycentric contraction operation, suitably defined at the level of quantile functions (see equation (2) for a precise definition). Intuitively, the model suggests that step $n + 1$ in the chain is obtained by pushing forward the n th step (‘shrunk’ slightly to allow for temporal stationarity) via

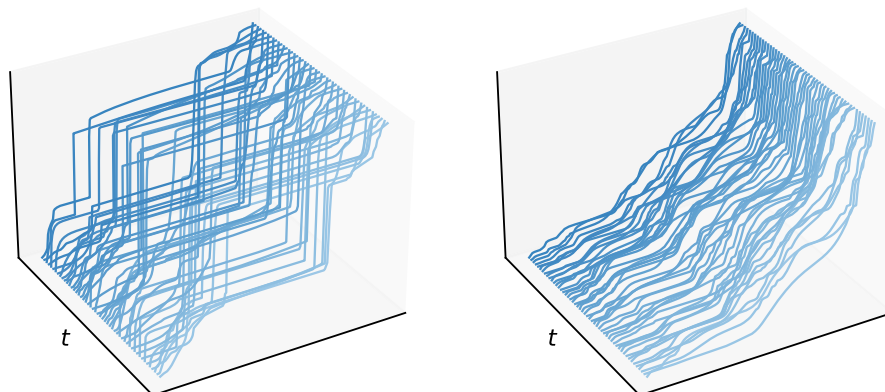


Figure 1. Examples of time series generated from equation (4) corresponding to two scenarios in Section 4. The left plot corresponds to map B with $\alpha = -0.9$, and the right plot corresponds to map D with $\alpha = 0.5$

a random perturbation of the deterministic deformation S . This is a direct autoregressive extension of Ghodrati and Panaretos (2022), employing the contractive device of Zhu and Müller (2021) to assure temporal stability in law. However, more modeling possibilities are available in our approach, and this is just the motivating one (see Section 3.2).

The rest of the article is organized as follows. After introducing some basic background and notation (Section 2), we revisit the problem of defining iterated random function systems of increasing maps. In particular, Section 3.1 presents a functional extension of the iterated system employed in Zhu and Müller (2021). This extension is then used in Section 3.2 in order to define three different possible notions of autoregression – in each case, the iterated transport map system serves to model a different characteristic of the distributional time series (e.g., the increments, the quantiles, or the generalized quantiles). We compare the resulting models to existing approaches in Section 3.3 and determine conditions for stationarity in Section 3.4. We then show in Section 3.5 that all three models can be fitted and analyzed using the same estimation theory – albeit applied to optimal maps that represent a different characteristic in each case. In particular, we establish identifiability, consistency, and rates of convergence. Finally, the finite sample performance of our methodology is illustrated on some simulated and real data (Sections 4 and 5). The proofs are collected in a separate Section, and we conclude with a discussion of some further possible generalizations.

2. BACKGROUND ON OPTIMAL TRANSPORT AND SOME NOTATION

In this section, we provide some background on optimal transport and associated probability metrics. For more background see, e.g., Panaretos and Zemel (2020). Let $\Omega = [\omega_0, \omega_1]$ be a closed interval of $\mathbb{R}$ and $\mathcal{W}_2(\Omega)$ be the set of Borel probability measures on Ω , with finite second moment. The 2-Wasserstein distance W between $\mu, \nu \in \mathcal{W}_2(\Omega)$ is defined by

$$d_{\mathcal{W}}(\nu, \mu) := \left(\inf_{\gamma \in \Gamma(\nu, \mu)} \int_{\Omega} |x - y|^2 d\gamma(x, y) \right)^{1/2},$$

where $\Gamma(\nu, \mu)$ is the set of couplings of μ and ν , i.e., the set of Borel probability measures on $\Omega \times \Omega$ with marginals ν and μ . It can be shown that $\mathcal{W}_2(\Omega)$ endowed with $d_{\mathcal{W}}^2$ is a metric space, which we simply call the Wasserstein space of distributions. A coupling γ is deterministic if it is the joint distribution of $\{X, T(X)\}$ for some deterministic map $T : \Omega \rightarrow \Omega$, called an optimal transport map. In such a case, we write $\nu = T\#\mu$ and say that T pushes μ forward to ν , i.e., $\nu(B) = \mu\{T^{-1}(B)\}$ for any Borel set B .

When the source distribution μ is absolutely continuous with respect to the Lebesgue measure, then the optimal coupling is induced by a non-decreasing map T , with the explicit expression $T = F_\nu^{-1} \circ F_\mu$, where F_ν^{-1} is the quantile function of ν , and F_μ is the cumulative distribution function of μ . It follows immediately that the composition of two optimal maps results in another optimal map.

In addition, we have the explicit expression

$$d_{\mathcal{W}}^2(\mu, \nu) = \int_0^1 \left| F_\mu^{-1}(p) - F_\nu^{-1}(p) \right|^2 dp. \quad (1)$$

Finally, we will use the notation $a \lesssim b$ to indicate that there exists a positive constant C for which $a \leq Cb$ holds (bounded above up to a universal constant). We denote by $\|\cdot\|_p$ the usual L^p norm of a function.

3. AUTOREGRESSIVE MODELS VIA ITERATED TRANSPORTATION

3.1. Random Iterated Transport

Our definition of autoregressive models for distributions will hinge on appropriately defined iterated random systems of transport maps (following the approach of Zhu and Müller (2021), to whom we compare below). This is a special case of a framework for studying questions about Markov chains via iterated random functions, going back to at least Diaconis and Freedman (1999). They define an iterated random function system on a state space $\mathcal{T}$ as

$$T_i = f(T_{i-1}; \theta_i),$$

for a family of transformations $\{f(\cdot; \theta) : \theta \in \Theta\}$ acting on $\mathcal{T}$, and random elements θ_i in some parameter space Θ , independent of $T_i \in \mathcal{T}$. By suitable choice of the family $f(\cdot; \theta)$ and some distribution on Θ they show how a plethora of Markov chains can be cast in this light.

In our case, the state space $\mathcal{T}$ will be the set of optimal transport maps

$$\mathcal{T} := \{T : \Omega \rightarrow \Omega \mid T(\omega_1) = \omega_1, T(\omega_2) = \omega_2, T \text{ is strictly increasing and continuous}\},$$

viewed as a closed and complete subset of the Lebesgue space $L^p(\Omega)$ equipped with the corresponding p -distance $\|\cdot\|_p$, for some $1 \leq p < \infty$ (we will mostly focus on $p = 2$). And, the question is how to define f and θ_i to generate an iterated random system that is sufficiently rich to serve as a basis for interesting autoregressive models, yet remains tractable and admits a non-degenerate stationary solution. Naively, one might simply posit that $\Theta = \mathcal{T}$ and $f(T; \theta) = \theta \circ T$, as increasing maps form a transformation group under composition. However, $f(\cdot; \theta)$ needs to be a contraction ‘on average’ (in a precise sense) for the Diaconis and Freedman (1999) results to be applicable.

This motivates forms of f that are ‘contractive compositions’. To this aim, given $|\alpha| \leq 1$, define the α -contraction of an optimal transport map to be the operator $T \mapsto [\alpha T]$ defined pointwise via

$$[\alpha T](x) = \begin{cases} x + \alpha(T(x) - x) & 0 < \alpha \leq 1 \\ x & \alpha = 0 \\ x + \alpha(x - T^{-1}(x)) & -1 \leq \alpha < 0. \end{cases} \quad (2)$$

This definition is due to Zhu and Müller (2021), under slightly different terminology/notation, and mimics the operation of contracting an unconstrained function by a scalar, but conforming to the constraints elicited by working in $\mathcal{T}$. Notice that $T \mapsto [\alpha T]$ is indeed a contraction on $\mathcal{T}$ with respect to L^1 norm, with the identity as its fixed point – any other fixed point must equal the identity almost everywhere by the Banach fixed-point theorem.

Finally, given $|\alpha| < 1$ and $\theta \in \mathcal{T}$ we can now make precise the notion of f being a ‘contractive composition’ map by defining

$$f(T; \theta) = \theta \circ [\alpha T].$$

To define an iterated random system, it suffices to put a probability distribution Q on $\mathcal{T}$, and make i.i.d. draws $\theta_i \sim Q$ yielding

$$T_i = f(T_{i-1}; \theta_i). \quad (3)$$

Our proposal is to draw i.i.d. elements of $\mathcal{T}$ with a specified expectation $S \in \mathcal{T}$, say $\theta_i = T_{\epsilon_i} \circ S$, for $\{T_{\epsilon_i}\}_{i=1}^N$ a collection of i.i.d random optimal maps satisfying $\mathbb{E}\{T_{\epsilon_i}(x)\} = x$ almost everywhere on Ω . Explicitly, our iteration is now

$$T_i = f(T_{i-1}; \underbrace{T_{\epsilon_i} \circ S}_{\theta_i}) = \underbrace{T_{\epsilon_i} \circ S}_{\theta_i} \circ [\alpha T_{i-1}]. \quad (4)$$

The degrees of freedom in this iteration are the choice of $S \in \mathcal{T}$ and $\alpha \in [-1, 1]$. In a statistical setting, these would be the targets of estimation. This definition extends the iteration of Zhu and Müller (2021) where S was a priori fixed to be the identity. Our extension seems natural and conceptually straightforward: it iterates contracted composition with perturbations of an arbitrary element of the transformation group, rather than with perturbations of the neutral element. Yet, it substantially complicates the subsequent probabilistic analysis and estimation theory. In exchange, we get a richer class of autoregressive models that exhibit advantages in the context of modeling and data analysis. We elaborate on the relationship and the nature of the extension in a subsequent paragraph. We then show that the iteration admits a unique stationary solution (under some additional assumptions). First, though, we explore how such an iterated random system of optimal maps could be used as a basis for distributional autoregression.

Remark 3.1. Consider the AR(1) model for scalar observations, denoted as X_i . The model can be expressed as:

$$X_i - \mu = \varphi(X_{i-1} - \mu) + \epsilon_i,$$

where ϵ_i represents the i.i.d. innovations with a mean of 0. A contraction toward the mean is achieved when the absolute value of φ is less than 1, that is, $|\varphi| < 1$. Our proposed AR(1) model for optimal maps seeks to approximate the optimal transport map at time $t = i$ using the transport map expressed as $S \circ [\alpha T_{i-1}]$. Notably, when $\alpha = 0$, the map T_i is approximated by S , which corresponds to a scenario where $\varphi = 0$.

3.2. Autoregressive Models

The main purpose of a random iteration (3) is the construction of a Markov chain model for a dependent sequence of probability distributions $\mu_i \in \mathcal{W}_2(\Omega)$, that will always be taken to possess a continuous cumulative distribution function. The models we seek are of autoregressive type, and so should ultimately be interpretable as a structural specification of the one-step conditional mean. Given stationary random sequence $\{T_i\}$ of optimal maps, there appear to be (at least) three different ways of doing so, by relating the T_i to some suitable feature of $\{\mu_i\}$:

- (I) Modeling the ‘increments’ $T_{\mu_i}^{\mu_{i-1}} := F_{\mu_i}^{-1} \circ F_{\mu_{i-1}}$ as being equal to T_i (we call these increments, as $T_{\mu_i}^{\mu_{i-1}}$ is the optimal map pushing μ_{i-1} forward to μ_i), or equivalently modeling the quantiles as

$$F_{\mu_i}^{-1} := T_i \circ F_{\mu_{i-1}}^{-1}.$$

When $\{T_i\}$ is stationary, this yields a process with stationary increments, but the process could be non-stationary (if so, it is interesting to understand if there is ‘drift’). This chain corresponds to specifying that the (usual) conditional expectation of $F_{\mu_i}^{-1}$ given $F_{\mu_{i-1}}^{-1}$ and $F_{\mu_{i-2}}^{-1}$ as

$$\mathbb{E}[F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1}, F_{\mu_{i-2}}^{-1}] = \mathbb{E}[F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1}, T_{i-1}] = \mathbb{E}\{f(T_{i-1}; \theta_i)\} \circ F_{\mu_{i-1}}^{-1} = \mathbb{E}\{\theta_i \circ [\alpha T_{i-1}]\} \circ F_{\mu_{i-1}}^{-1}.$$

The precise form of $\mathbb{E}[T_i]$ will depend on the stationary solution of $T_i = f(T_{i-1}; \theta_i)$.

(UQ) Modeling the (uniform) quantiles $F_{\mu_i}^{-1}$ as being equal to T_i ,

$$F_{\mu_i}^{-1} := T_i.$$

This automatically yields a stationary process when $\{T_i\}$ is stationary, directly interpretable at the level of quantiles, and corresponds to specifying the (usual) conditional expectation of $F_{\mu_i}^{-1}$ given $F_{\mu_{i-1}}^{-1}$ as

$$\mathbb{E}[F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1}] = (\mathbb{E}\theta_i) \circ [\alpha F_{\mu_{i-1}}^{-1}] = S \circ [\alpha F_{\mu_{i-1}}^{-1}] = f(F_{\mu_{i-1}}^{-1}; S).$$

This model corresponds to an autoregressive extension of the model in Ghodrati and Panaretos (2022).

(GQ) Modeling the generalized quantiles (Chernozhukov *et al.*, 2017) or μ -quantiles $F_{\mu_i}^{-1} \circ F_{\mu}$ with respect to a known measure μ as being equal to T_i . This also immediately yields stationarity and (under regularity conditions) is equivalent to stating $\mu_i = T_i \# \mu$, in effect modeling the μ_i as serially dependent ‘perturbations’ of a fixed μ . This corresponds to specifying the (usual) conditional expectation of $F_{\mu_i}^{-1}$ given $F_{\mu_{i-1}}^{-1}$ as

$$\mathbb{E}[F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1}] = (\mathbb{E}\theta_i) \circ [\alpha [F_{\mu_{i-1}}^{-1} \circ F_{\mu}]] = S \circ [\alpha [F_{\mu_{i-1}}^{-1} \circ F_{\mu}]] = f(F_{\mu_{i-1}}^{-1} \circ F_{\mu}; S).$$

In the special case where μ is the uniform distribution, this model reduces to (UQ).

Note that setting $\alpha = 1$ in (UQ) yields the same model as setting $\alpha = 0$ in (I), interpretable as a random walk, and this we shall revisit. In Section 5 we will focus on (UQ) and (I) to model sequential distributional data and discuss the merits/drawbacks of each approach. Model (GQ) can actually be seen to be a variant of the model (UQ) albeit under a modification of the definition of the contraction operator itself – see Section 6.2, and especially Remark (6.14) for an equivalent characterization of the model (GQ).

Remark 3.2. In practice, when selecting between models (I) and (UQ), it may be beneficial to recognize that model (I) is predicated on the assumption of stationary consecutive maps, while (UQ) relies on the assumption of stationary distribution series. Visual inspection of both maps and quantile functions for stationarity is feasible, in addition to implementing various tests for stationarity in functional time series, as highlighted in recent literature (e.g., Aue and Van Delft, 2020; van Delft *et al.*, 2021). However, it is important to note that it can very well happen that *both* the consecutive maps (increments) *and* the distributions themselves be stationary. This is because a stationary sequence of distributions will necessarily have stationary increments. But the converse is obviously not necessarily true. Such devices, therefore, might rule out (UQ) in favour of (I), but cannot do the converse.

3.3. Comparison with Related Work

Our iteration (4) represents a generalization of the iteration in Zhu and Müller (2021), by combining their notion of α -contraction (which they call *distributional scalar multiplication*), with the functional structure of the model in Ghodrati and Panaretos (2022). Specifically, Zhu and Müller (2021) considered autoregressive models for distributional time series, based on the iterative system of optimal transport maps

$$T_i = T_{\epsilon_i} \circ [\alpha T_{i-1}]. \quad (5)$$

That this is a special case of our system (4) when S is fixed to be the identity map $\text{id}(x) = x$. Their clever α -contraction, combined with classical results on iterated random function theory, allows one to deduce the existence of a unique stationary solution to the iteration (5) thanks to the contracting effect of α for $-1 < \alpha < 1$ (and some additional technical assumptions).

However, basing a distributional autoregressive model on this system is restrictive in two important ways:

1. As a stochastic model, the system (5) is parametric and univariate: the only unknown is the scalar coefficient $\alpha \in (-1, 1)$. Correspondingly, when basing our model on that iteration (with any of the three interpretations specified in the previous section), the temporal dependence of μ_i on μ_{i-1} will be completely specified up to an unknown scalar parameter. This is reminiscent of autoregressive models on the real line but is arguably overly restrictive in a functional data analysis (or even multi-variate analysis) setting, where the temporal dependence is very likely more complex. A genuinely functional model would replace the scalar coefficient with a suitable *functional coefficient*, e.g., a nonlinear operator.
2. If a stationary solution to system (5) exists, then it must satisfy $\mathbb{E}(T_i) = \text{id}$. To see this, recall the definition of the scalar multiplication (2) and observe that

$$\mathbb{E}[T_i] = \mathbb{E}[T_{i+1}] = \mathbb{E}[\mathbb{E}[T_{i+1}|T_i]] = \mathbb{E}[\alpha T_i].$$

This is consequential if using the sequence T_i to induce a distributional time series $\{\mu_i\}$. In the (I) model, where T_i models the increments between consecutive μ_i , this implies that the conditional Fréchet mean (in the Wasserstein metric) of μ_i given μ_{i-1} is exactly equal to μ_{i-1} , a sort of ‘Fréchet martingale’. Effectively this trivializes the regressor relationship to be an identity – there is no modeling flexibility for the conditional mean, only the conditional variance (via α). In the (UQ) model, where $T_i \equiv F_{\mu_i}^{-1}$ is taken as the quantile function of μ_i , the fact that $\mathbb{E}(T_i) = \text{id}$ implies that the distributional autoregression model can only admit the uniform distribution as its Fréchet mean (with respect to the Wasserstein metric). There is no flexibility in the modeling of the marginal mean.

By contrast, models based on our system (4) are genuinely functional, since on account of the unknown transport map S . Furthermore, our model can accommodate *any* distribution as its Fréchet mean: given any optimal map $T \in \mathcal{T}$, there exist S and α such that $\mathbb{E}(T_i) = T$.

The optimal map interpretation of our system (4) is an autoregressive modification of the distributional optimal transport regression model of Ghodrati and Panaretos (2022). Ghodrati and Panaretos (2022) define the regression model

$$v_i = T_{\epsilon_i} \# (S \# \mu_i), \quad \{\mu_i, v_i\}_{i=1}^N,$$

where $S : \Omega \rightarrow \mathbb{R}$ is an unknown optimal map and $\{T_{\epsilon_i}\}_{i=1}^N$ is a collection of i.i.d random optimal maps satisfying $\mathbb{E}\{T_{\epsilon_i}(x)\} = x$ almost everywhere on Ω . By direct analogy, an autoregressive model (optimal map interpretation) for a time series of distributions $\{\mu_i\}$ would be defined as

$$\mu_i = T_{\epsilon_i} \# (S \# \mu_{i-1}), \quad (6)$$

which is equivalent to model (4) when $\alpha = 0$ and when we interpret T_i such that $\mu_i = T_i \# \mu_{i-1}$, i.e., the optimal map interpretation. If we take the quantile interpretation, the two models are again related for $\alpha = 1$ since model (4) is equivalent to

$$F_i^{-1} = T_{\epsilon_i} \circ S \circ F_{i-1}^{-1}.$$

However, assuming the noise maps T_{ϵ_i} are close to identity, and focus on the influence of the map S , a pattern emerges. For values situated between two of S 's fixed points, their eventual trajectory is dictated by S 's position relative to the identity. If S lies above the identity line, these values trend toward the higher fixed point. Conversely, if S is below the identity line, they move toward the lower fixed point. As a result, over time, the sequence of cumulative distribution functions F_{i-1}^{-1} begins to resemble a step function. The transition points in this function align with the fixed points of S . Consequently, the associated distribution, μ_i , tends to fluctuate around a composition of Dirac measures. This is where we combine the functional structure of Ghodrati and Panaretos (2022) with the scalar 'contractive effect' introduced by Zhu and Müller (2021) – intuitively, the magnitude of α regulates the non-degeneracy of the model. The next section demonstrates that this combined extension does indeed yield a unique stationary solution.

3.4. Existence of Unique Stationary Solution

We now turn to establish the existence of a unique stationary solution for the system (4). We will use the results of Wu and Shao (2004), extending to our iteration (4) the steps follows by Zhu and Müller (2021) in the context of iteration (5). Let $\{T_{\epsilon_i}\}_{i=1}^N$ be a collection of i.i.d random optimal maps satisfying $\mathbb{E}\{T_{\epsilon_i}(x)\} = x$ almost everywhere on Ω . Define $\Phi_i, \tilde{\Phi}_{i,m} : \mathcal{T} \rightarrow \mathcal{T}$ by

$$\begin{aligned}\Phi_i(T) &= f(T; T_{\epsilon_i} \circ S) = T_{\epsilon_i} \circ S \circ [\alpha T] \\ \tilde{\Phi}_{i,m}(T) &= \Phi_i \circ \Phi_{i-1} \circ \cdots \circ \Phi_{i-m+1}(T).\end{aligned}\quad (7)$$

The Assumption 3.3 stipulates.

Assumption 3.3. (Moment Contracting Condition (Wu and Shao, 2004)). Suppose there exists $\eta > 0, Q_0 \in \mathcal{T}, C > 0$ and $r \in (0, 1)$ such that

$$\mathbb{E}\|\tilde{\Phi}_{i,m}(Q_0) - \tilde{\Phi}_{i,m}(T)\|_2^\eta \leq Cr^m \|Q_0 - T\|_2^\eta, \quad (8)$$

holds for all $i \in \mathbb{Z}, m \in \mathbb{N}$ and all $T \in \mathcal{T}$.

Lemma 3.4. Assume the parameters of the model (4) satisfy the Assumption 3.3. Then for all $T \in \mathcal{T}$, $\tilde{T}_i := \lim_{m \rightarrow \infty} \tilde{\Phi}_{i,m}(T) \in \mathcal{T}$ (convergence in L^2 sense) exists almost surely and does not depend on T . In addition, $\tilde{T}_i$ is a stationary solution to the following system of stochastic transport equations:

$$T_i = T_{\epsilon_i} \circ S \circ [\alpha T_{i-1}], \quad i \in \mathbb{Z},$$

and is unique almost surely.

Remark 3.5. Zhu and Müller (2021) proposed a specific parameter condition for their model that ensures Assumption 3.3 is satisfied. We provide a similar sufficient condition for the parameters of Model (4) that also guarantees the satisfaction of Assumption 3.3. Let L_ϵ be constant such that $\mathbb{E}|T_\epsilon(x) - T_\epsilon(y)|^2 \leq L_\epsilon^2 |x - y|^2$. Assuming $\alpha \geq 0$, if $|S(x) - S(y)| \leq L_S |x - y|$ and $\alpha L_S L_\epsilon < 1$, then Model (4) satisfies Assumption 3.3 with $\eta = 2$ and $r = \sqrt{\alpha L_S L_\epsilon}$. Similarly, if $\alpha < 0$, suppose the aforementioned conditions are met and define $\mathcal{T}_{l,u} = \{T \in \mathcal{T} : 0 < L_l \leq T' \leq L_u < \infty\}$ and assume $\{T_i\} \subset \mathcal{T}_{l,u} \subset \mathcal{T}$ (see Lemma 6.2 and Proposition 6.5). Then Model (4) also satisfies Assumption 3.3 with $\eta = 2$ and $r = \sqrt{\alpha L_S L_\epsilon}$.

3.5. Estimation and Statistical Analysis

We consider a time series of continuous distributions $\mu_i \in \mathcal{W}_2(\Omega)$ and corresponding time series $T_i \in \mathcal{T}$, which are related by one of the models from Section 3.2. Although the methods to obtain T_i may differ for each model,

we can always obtain T_i by observing μ_i . Our analysis is thus applicable to all three models studied, but in each different model, the T_i will represent a different feature of the distributional time series. For the remainder of our analysis, we assume that T_i is a (the) stationary solution obtained from system (4).

As discussed in Section 3.3, when S is fixed a priori to be the identity, our iteration (4) will reduce to that of Zhu and Müller (2021). In this simplified setting, Zhu and Müller (2021) use the fact that α is the minimizer of $\mathbb{E}\|T_{i+1} - [\alpha T_i]\|_2^2$ to obtain a closed form expression for α as

$$\frac{\int_{\Omega} \mathbb{E}[(T_{i+1}(x) - x)(T_i(x) - x)]dx}{\int_{\Omega} \mathbb{E}[(T_i(x) - x)^2]dx},$$

when $\alpha \in [0, 1)$ or

$$\frac{\int_{\Omega} \mathbb{E}[(T_{i+1}(x) - x)(x - T_i^{-1}(x))]dx}{\int_{\Omega} \mathbb{E}[(x - T_i^{-1}(x))^2]dx},$$

when $\alpha \in (-1, 0)$. These show that α can be interpreted as the autocorrelation coefficient, and can be estimated by its empirical version, which allows for a straightforward path to consistency and parametric rates of convergence.

However, our more general iteration (4), involves an arbitrary non-decreasing map S that also needs to be estimated. Consequently, not only are there no closed forms for the estimands (α, S) but the estimation problem becomes distinctly non-linear.

To motivate our estimators, we note that if S were known, then α could be estimated by nonlinear least squares, as the minimizer of $\frac{1}{N} \sum_{i=1}^N \|S \circ [\alpha T_{i-1}] - T_i\|_2^2$. On the other hand, if α were known, then a natural candidate to estimate S would be the ergodic average

$$S_{N,\alpha} := \frac{1}{N} \sum_{j=1}^N T_j \circ [\alpha T_{j-1}]^{-1}.$$

This is because the definition of the iteration $T_j = f(T_{j-1}; T_{e_j} \circ S) = T_{e_j} \circ S \circ [\alpha T_{j-1}]$, combined with the assumption that $\mathbb{E}[T_{e_j}(x)] = x$, yields that

$$\mathbb{E}\{T_j \circ [\alpha T_{j-1}]^{-1}\} = \mathbb{E}\{T_{e_j} \circ S\} = S.$$

Since $S_{N,\alpha}$ is available in closed form for any choice of α , this suggests plugging the expression $S_{N,\alpha}$ for S into the sum of squares, to obtain an objective that depends only on α . Minimizing the said objective over α one obtains an estimator $\hat{\alpha}$, which automatically induces an estimator of S in the form of $S_{N,\hat{\alpha}}$.

Formally, we define the estimators $(\hat{\alpha}_N, S_{N,\hat{\alpha}_N})$ of (α, S) as follows:

$$\hat{\alpha}_N := \arg \min_{\alpha} M_N(\alpha), \quad (9)$$

where

$$\begin{aligned} M_N(\alpha) &:= \frac{1}{N} \sum_{i=1}^N g_{\alpha}(T_{i-1}, T_i, S_{N,\alpha}) \\ g_{\alpha}(T_{i-1}, T_i, S) &:= \|S \circ [\alpha T_{i-1}] - T_i\|_2^2 \\ S_{N,\alpha} &:= \frac{1}{N} \sum_{j=1}^N T_j \circ [\alpha T_{j-1}]^{-1}. \end{aligned} \quad (10)$$

To analyze the behavior of our estimators, we also define the following population quantities:

$$\begin{aligned} S_\alpha &:= \mathbb{E}[T_j \circ [\alpha T_{j-1}]^{-1}] \\ M(\alpha) &:= \mathbb{E}g_\alpha(T_{i-1}, T_i, S_\alpha). \end{aligned} \quad (11)$$

The left-hand sides do not depend on j due to stationarity, which will be assumed throughout.

For the sake of clarity, we will henceforth denote the true parameters of the model using boldface fonts, namely as $(\alpha, \mathbf{S})$.

Theorem 3.6. If the true parameters of the model are $(\alpha, \mathbf{S})$, then $S_\alpha = \mathbf{S}$.

Proof. For the true α , we have

$$\begin{aligned} S_\alpha &= \mathbb{E}[T_j \circ [\alpha T_{j-1}]^{-1}] \\ &= \mathbb{E}[T_{e_j} \circ \mathbf{S} \circ [\alpha T_{j-1}] \circ [\alpha T_{j-1}]^{-1}] \\ &= \mathbb{E}[T_{e_j} \circ \mathbf{S}] = \mathbf{S}. \end{aligned} \quad (12)$$

We show the consistency of the estimators $(\hat{\alpha}_N, S_{N,\hat{\alpha}_N})$ in the following four steps corresponding to the Lemmas 3.7, 3.8, 3.9 and Theorem 3.10 respectively:

- α is the unique minimizer of $M(\alpha)$.
- $S_{N,\alpha}$ converges uniformly (with respect to α) in probability to S_α in L^2 .
- $M_N(\alpha)$ converges uniformly in probability to $M(\alpha)$.
- we conclude the consistency (and identifiability) using the M-estimation theory.

Lemma 3.7. [Unique Minimizer of $M(\alpha)$] For any $\alpha \neq \alpha$ we have

$$M(\alpha) = \mathbb{E}g_\alpha(T_{i-1}, T_i, S_\alpha) < \mathbb{E}g_\alpha(T_{i-1}, T_i, S_\alpha) = M(\alpha),$$

where α is the true α .

Now we show that $S_{N,\alpha}$ converges to S_α in probability for any α and also prove a central limit theorem (CLT) for $S_{N,\alpha}$.

If $\alpha = \alpha$, then it is straightforward to argue that $S_{N,\alpha}$ converges to S_α : first note that for any $x \in [0, 1]$,

$$S_{N,\alpha}(x) = \frac{1}{N} \sum_{j=1}^N T_j \circ [\alpha T_{j-1}]^{-1}(x) = \frac{1}{N} \sum_{j=1}^N T_{e_j} \circ \mathbf{S}(x).$$

Since the terms in the expression are i.i.d with mean $\mathbf{S}$, the strong law of large numbers yields that $S_{N,\alpha}(x)$ converges to $S_\alpha(x)$. From Theorem 3.6, we know that the true $\mathbf{S} = S_\alpha$. Therefore in this case that $\alpha = \alpha$, $S_{N,\alpha}$ converges in probability to $S_\alpha = \mathbf{S}$. However, in general, when $\alpha \neq \alpha$ the terms $T_j \circ [\alpha T_{j-1}]^{-1}$ are not independent for different j . Therefore, we first show that since $\{T_j\}$ satisfies the moment generating condition (see Lemma 6.1 in the appendix), we can quantify the dependency between the terms in the sequence $T_j \circ [\alpha T_{j-1}]^{-1}$ and apply CLT methods developed for functional time series.

Lemma 3.8. (Central limit for $S_{N,\alpha}$). Suppose the parameters of the iteration (4) satisfy the Assumption 3.3. Then for any α , there is a Gaussian process Γ_α such that

$$\sqrt{N}(S_{N,\alpha} - S_\alpha) \xrightarrow{d} \Gamma_\alpha, \quad \text{in } L^2.$$

Also,

$$\sup_{\alpha} \|S_{N,\alpha} - S_{\alpha}\|_2 = o_{\mathbb{P}}(1).$$

Lemma 3.9. Suppose the parameters of the iteration (4) satisfy the Assumption 3.3. Then for any α , there is a $\sigma_{\alpha} \geq 0$ such that

$$\sqrt{N}[M_N(\alpha) - M(\alpha)] \rightarrow N(0, \sigma_{\alpha}^2).$$

Moreover,

$$\sup_{\alpha} |M_N(\alpha) - M(\alpha)| = o_{\mathbb{P}}(1).$$

Theorem 3.10. (Identifiability and Consistency). Under Assumption 3.3, the parameters of the iteration (4) are identifiable and $(\hat{\alpha}_N, S_{N,\hat{\alpha}_N})$ are consistent estimators for $(\alpha, \mathbf{S})$.

Theorem 3.11. (Rate of Convergence). Let $\mathcal{T}_{l,u} = \{T \in \mathcal{T} : 0 < L_l \leq T' \leq L_u < \infty\}$ and suppose $\{T_i\} \subset \mathcal{T}_{l,u} \subset \mathcal{T}$. Under Assumption 3.3 and twice differentiability of the T_i , we have

$$N^{\frac{1}{2}} |\hat{\alpha}_N - \alpha| = O_{\mathbb{P}}(1),$$

$$N^{\frac{1}{2}} \|S_{N,\hat{\alpha}_N} - \mathbf{S}\|_2 = O_{\mathbb{P}}(1).$$

4. SIMULATION EXPERIMENTS

In this section, we probe the behavior of our models, and the finite sample performance of our estimation framework, via simulation. To generate the noise maps T_{ϵ_i} , we use a class of random optimal maps introduced in Ghodrati and Panaretos (2022) that are modifications of the maps used in Panaretos and Zemel (2016): Let K be a random integer with a symmetric distribution around zero. We define $\zeta_K : [0, 1] \rightarrow [0, 1]$ by

$$\zeta_0(x) = x, \quad \zeta_K(x) = x - \frac{\sin(\pi Kx)}{|K|\pi}, \quad K \in \mathbb{Z} \setminus \{0\}. \quad (13)$$

These are strictly increasing smooth functions satisfying $\zeta_K(0) = 0$ and $\zeta_K(1) = 1$. For $x \in [0, 1]$ we have $\mathbb{E}[\zeta_K(x)] = x$, as required in the definition of model (4). The random maps T_{ϵ} will be a mixture of the maps (13) as defined in Ghodrati and Panaretos (2022).

Each plot in Figure 2 corresponds to a time series simulation with a different combination of $\mathbf{S}$ and α . Each column corresponds to a different value of $\alpha \in \{-0.9, -0.5, 0, 0.5, 0.9\}$ from left to right. In the three top rows, $\mathbf{S}$ is chosen to be ζ_K for $K = \{-6, -4, -2\}$ from top to bottom. In row four, $\mathbf{S}$ is the average of ζ_1 and an instance of T_{ϵ} . Rows five and six exemplify the method on non-differentiable and discontinuous maps $\mathbf{S}$ respectively.

Plots that fall within the bounding red rectangle correspond to settings where our theory is guaranteed to apply. Plots outside of that rectangle are not guaranteed to be covered by our theory: they either distinctly violate our assumptions (such as the last row where the true map is not continuous, as required) or we cannot confirm whether the Assumption 3.4 holds true. Starting from the identity map, we generate a time series with 300 iterations and discard the first 100 maps of the series. The remaining 200 maps $\{T_i\}$ are shown in light blue, the true map $\mathbf{S}$ is in black, and the estimated map is in orange. For each time series, we show the estimated $\hat{\alpha}$ and the error between the estimator and true map in $\|\cdot\|_2$ -norm.

As expected from Remark 3.5, smaller values of $|\alpha|$ lead to time series which apparently oscillate around the mean of the stationary time series, which in turn leads to the convergence of our estimator with respect to the true

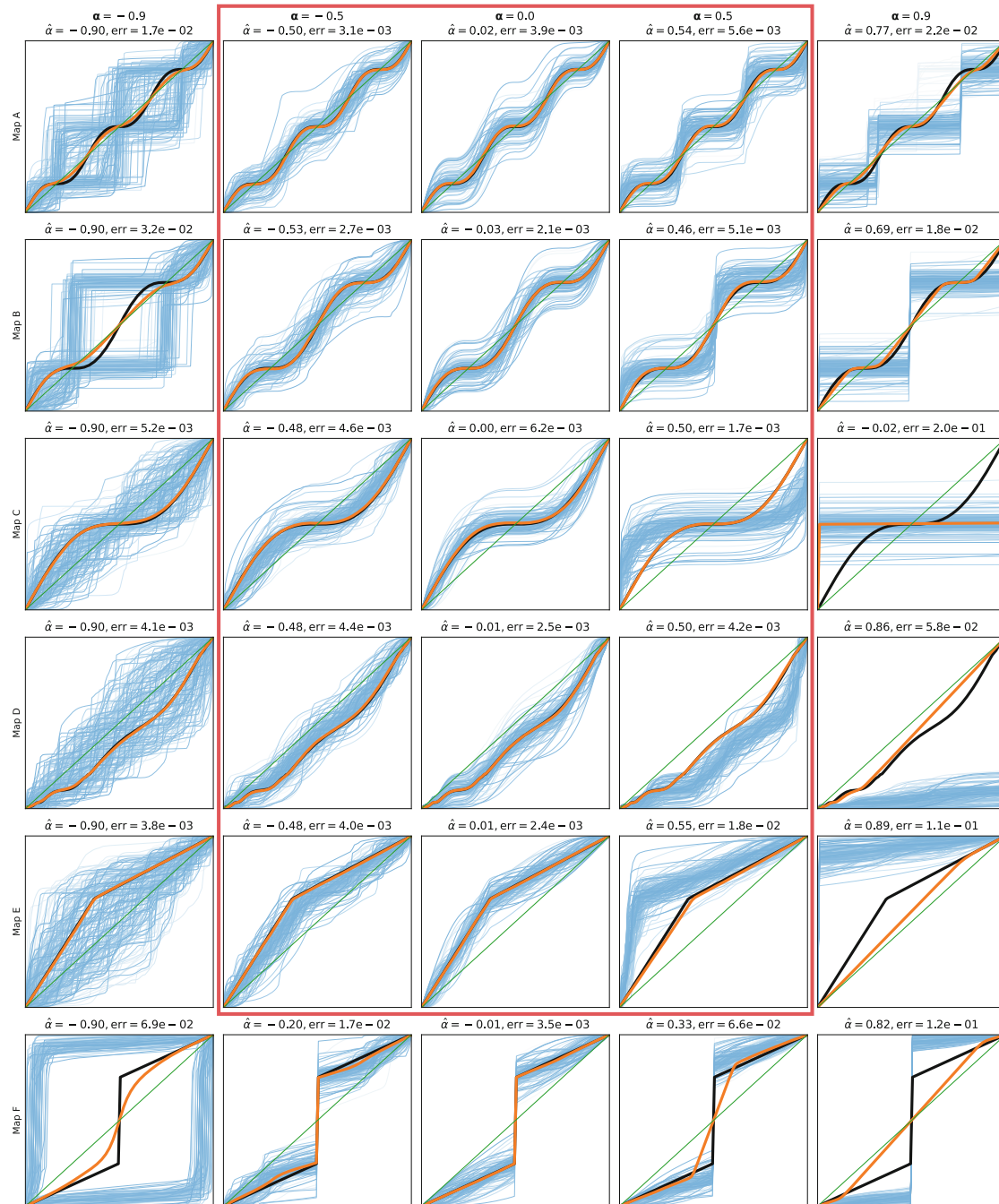


Figure 2. Estimated map (orange) versus the true map (black) for different combinations of α and S . The blue line denotes a simulated time series of length 300 (with the initial 100 maps omitted), while the green line represents the id map. Each column corresponds to a different value of α , ranging from -0.9 on the left to 0.9 on the right. The top three rows show results for $S = \zeta_K$ where K is chosen from $\{-6, -4, -2\}$ from top to bottom. In the fourth row, S is the average of ζ_1 and an instance of T_e . The fifth and sixth rows demonstrate the method on non-differentiable and discontinuous maps respectively. The cases within the red rectangle are covered by our theoretical guarantees.

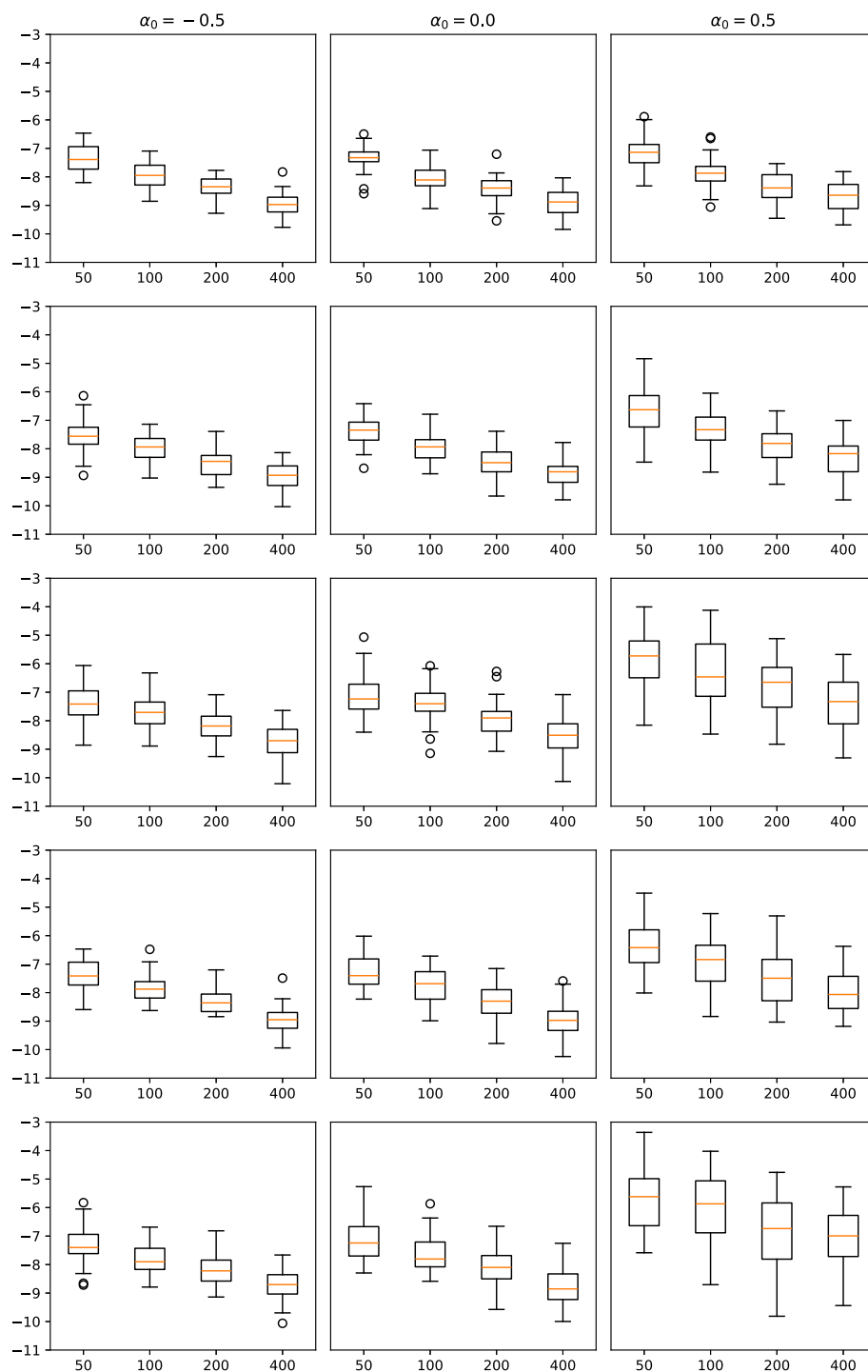


Figure 3. Detailed analysis of the cases highlighted by the red rectangle in Figure 2. The rows in this figure align with its top five rows. Each boxplot illustrates the impact of the time series length (X-axis) on the estimation error in log scale (Y-axis) for the corresponding map $\mathbf{S}$. Each scenario is simulated 50 times

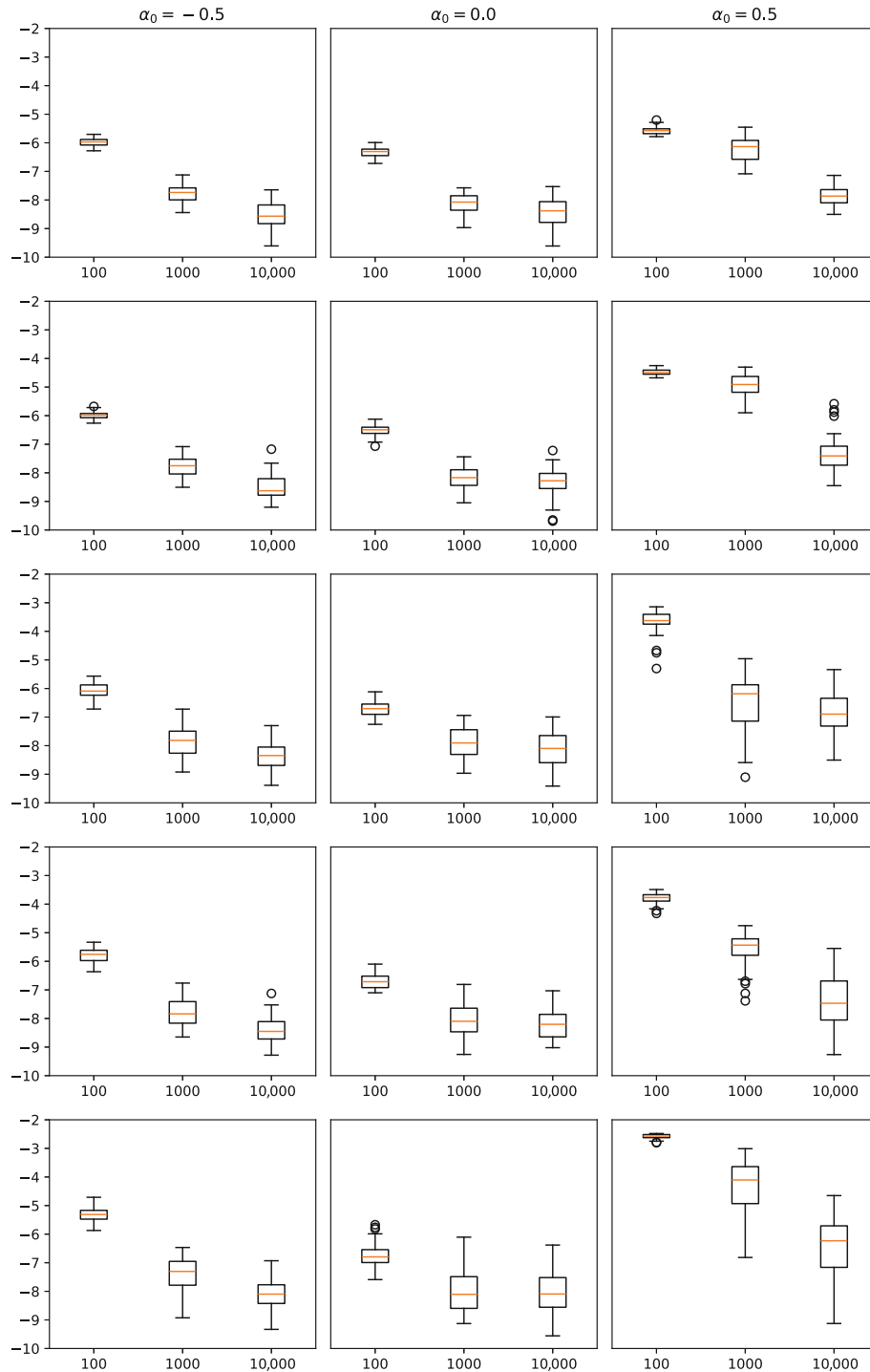


Figure 4. Detailed analysis of the cases highlighted by the red rectangle in Figure 2. The rows here correspond to the first five rows of the same figure. In this (UQ) model, rather than utilizing direct quantiles, empirical quantiles are employed for parameter estimation. Each boxplot illustrates the relationship between the sample size used for empirical quantile computation (X-axis) and the estimation error of the map $\mathbf{S}$ in log scale (Y-axis). All configurations were replicated 50 times for simulation

map. In particular, good agreement is seen between the estimator and true map for values of $|\alpha|$ up to 0.5 at least, only noticeably failing for $\alpha = 0.5$ in the discontinuous map case (where our theoretical guarantee does not apply due to the discontinuity).

Larger values of $|\alpha|$ can still lead to similar stationary state time series (sometimes even outside of the red rectangle, where our theoretical guarantees apply) but with naturally larger oscillations. Still, a good agreement between the estimator and ground truth is observed. This can depend on the choice of map $\mathbf{S}$ and the precise value of α . For instance, in the third, fourth, and fifth rows, when $\alpha = -0.9$. In the remaining rows of the first column, the stationary state behavior changes to a period-two time series (with noise) where the maps oscillate alternatively between two maps related by inversion (recall that negative values of α imply an inversion of the map T_{i-1} at each time step). Nevertheless, the estimator is able to capture features of the $\mathbf{S}$ map that are not visible in the time series itself: notably, the discontinuous step in row six is present in the estimated map.

In the other extreme of $\alpha = 0.9$, the time series maps are close to step-like functions with some variation in the step height. The maps are in fact still oscillating around the mean of the stationary time series that is very close to the step-like map $\mathbf{S}^\infty$, which is the mean of the solution to the model (4) when $\alpha \rightarrow 1$, that is $T_s = \mathbf{S} \circ T_s$. However, the performance of the estimator is the worst in this limit.

Do note that the family of maps $\zeta_K(x)$ is not symmetric with respect to inversion in the sense that the derivative of $\zeta_K(x)$ is 0 at some fixed points ($\zeta_K(x) = x$) but is never infinite, and therefore the random maps T_e , which are derived from $\zeta_K(x)$, are biased in this way. For this reason, the vertical variance observed in most maps is much more pronounced than the horizontal one, which is very clear in the case $\alpha = 0.9$.

In complement to Figure 2, we provide an in-depth analysis in Figure 3, focusing on the scenarios encapsulated by the red rectangle. This figure emphasizes the impact of time series length on the estimation error of map $\mathbf{S}$. The boxplots clearly illustrate the variations in estimation errors for time series lengths of $\{50, 100, 200, 400\}$. Notably, the median values of these boxplots approximately align along a straight line with a slope of roughly -0.05 . Hence, the simulation is aligned with Theorem 3.11, which posits that the estimation error scales with the length of the observed time series N at a rate proportional to $N^{-1/2}$. Each configuration has been simulated 50 times to construct these boxplots.

Expanding on our analysis, Figure 4 examines scenarios where the time series of distributions are not directly observed but estimated through samples. Specifically, this focuses on the (UQ) model, wherein empirical quantiles replace direct quantiles for parameter estimation. Each boxplot corresponds to one of the parameter settings within the red rectangle of Figure 2, with the sample size for quantile estimation taken from the set $\{100, 1000, 10,000\}$. The plots demonstrate that in every scenario, increasing the sample size leads to a reduction in error. However, when $\alpha > 0$, a larger sample size is essential to obtain a reliable estimate for the map $\mathbf{S}$.

5. ILLUSTRATIVE DATA ANALYSIS

In this section, we consider the distribution of minimum daily temperatures recorded in the summer of the years from 1960 to 2020 from several airports in the USA (available at <https://www.ncei.noaa.gov>). That is, the years are taken as the time index, and for any given time index we observe a distribution over the temperature scale (representing the distribution of minimal temperatures over that year's summer). Thus, each airport gives rise to a distributional time series. This dataset has been also analyzed by Zhu and Müller (2021) to demonstrate their own distributional autoregressive model, which allows for constructive comparison.

We examine the daily minimum temperature for June, July, August, and September from 1960 to 2020 in four locations: Chicago O'Hare International Airport, Atlanta Hartsfield-Jackson International Airport, Phoenix Airport, and New Orleans Airport. The corresponding distributions are displayed in Figure 5.

The map sequence elicited by adopting the increment model (Model (I)) is shown in Figure 6(a). These maps are obtained by calculating the optimal maps between consecutive annual temperature distributions for each location. These maps exhibit oscillations around the identity, except in the subdomains corresponding to extreme temperature values. In the lower extreme, the maps impose a cutoff on the lower end of the support of the temperature

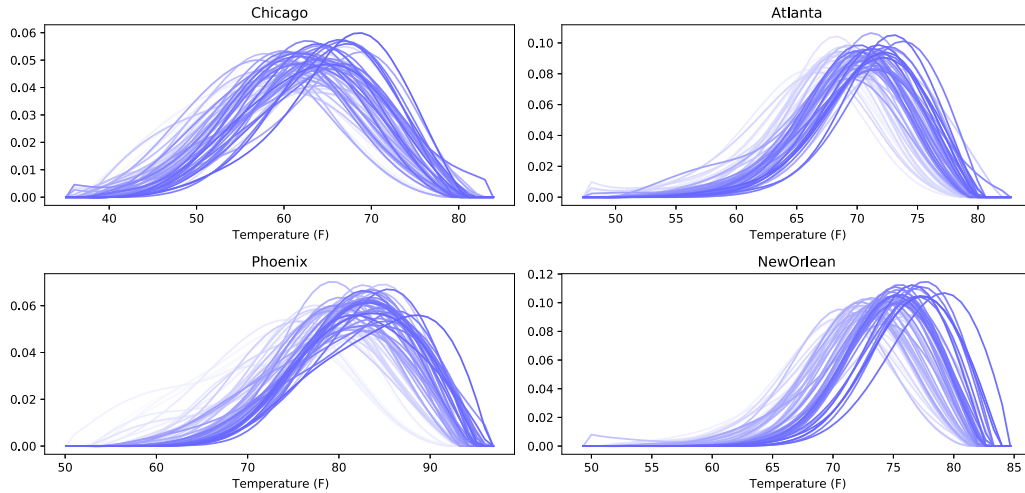


Figure 5. Time series of distribution of daily minimum temperature in summer from 1960 to 2020 at Chicago Ohare international airport, Atlanta Hartsfield Jackson international airport, Phoenix airport, and New Orleans airport. The shading reflects the time index: the fainter the curve, the earlier in time it corresponds to

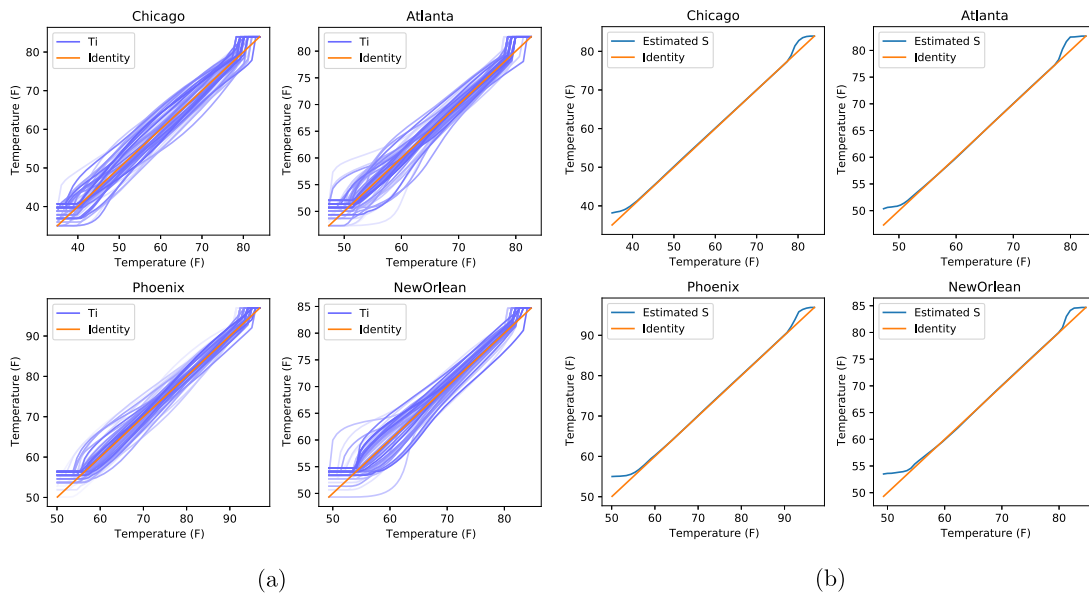


Figure 6. (a) Time series $\{T_i\}$ (blue) based on model (I) and identity map (orange) for the four locations. (b) Estimated map S (blue) and identity map (orange). Faint blue shading corresponds to early years, and bold shading to later years

distribution, while the higher end is pushed toward higher values and eventually reaches the extreme of the support. This implies that extreme temperatures are increasing, indicating that the coldest and hottest nights in summer are becoming hotter.

Figure 6(b) presents the estimates of S obtained using Model (I), where the estimated α was found to be 0 up to three decimal points for all airports. This suggests that the optimal maps T_i are independent from each other and, on average, they are equal to the estimated maps S presented. The estimated S maps are very similar across

all airports, effectively being the identity map in the middle portion of the support and above the identity at the extreme points.

Examining the maps generated by fitting Model (I), i.e., computing the optimal maps between consecutive annual distributions in Figure 6(a), we can observe an increasing trend in the cutoff value of the lower endpoint over time. This implies that the time series of optimal maps may not be stationary. Of course, the S maps are not able to capture the overall increase in the cutoff value of the lower end over time: the plateaus of the S maps are just the averages of the optimal maps T_i and don't show this trend. Indeed, a problem of modeling such data is that the system may be dynamically evolving due to factors like global warming, and it is not obvious a priori if stationary regimes exist that can be captured by our models.

However, using the uniform quantile model (Model (UQ)), the resulting maps are more interpretable and reveal more refined dynamics beyond the cutoffs at the extremes. To obtain these maps, we fitted iteration (4) to the time series of quantile functions of the temperature distributions. The quantile functions are shown in Figure 7(a). The resulting estimated maps S are in Figure 7(b), and the estimated $\hat{\alpha}$ for the four airports are $\{0.39, 0.80, 0.89, 0.89\}$. All the maps show a cutoff at the lower end and a fixed point in the second half of the support where the derivative is smaller than 1. The fixed point implies a point of stability, and the derivative indicates a trend toward a concentration of mass around this point. For example, when neglecting the effect of the noise map, if at a certain time the time series is at a distribution concentrated away from the fixed point, the subsequent distributions will gradually progress toward a distribution concentrated closer to the fixed point. Again, the model may be failing to capture a trend of ever-increasing temperature, or it might imply a stabilization at temperatures indicated by the fixed points, which could become the new norm.

Even if the model is possibly misspecified, the estimated maps S are still able to condense several features of the time series of distributions. Namely, the reduction of extreme cold events and the progression toward higher modal temperatures which may or may not be static.

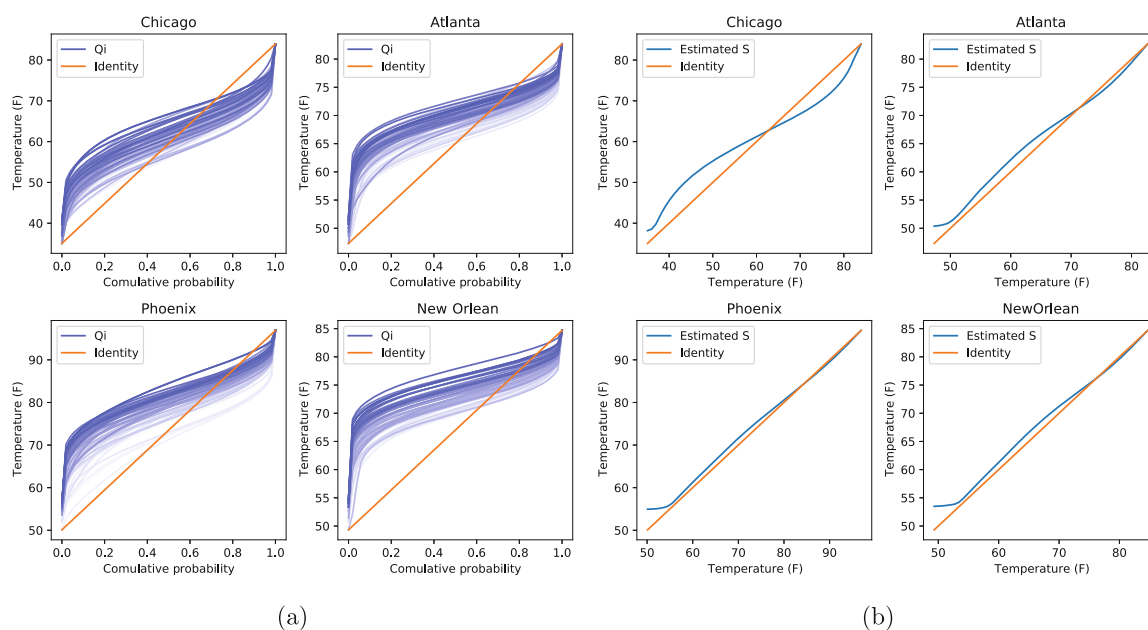


Figure 7. (a) Time series of quantile functions (blue) based on model (UQ) and identity map (orange) for the four locations. (b) Estimated map S (blue) and identity map (orange). Faint blue shading corresponds to early years, and bold shading to later years

There is an interesting observation to be made given that the estimated α when fitting the intercept model (I) is numerically 0 while it is in (0,1) when fitting the quantile model (UQ). Specifically, in combination, these results suggest that the quantile model is, in a certain sense, a better fit to the data. The reasoning is as follows. Recall that the increment model (I) with $\alpha = 0$ is equivalent to the quantile model (UQ) when $\alpha = 1$, and corresponds to ‘trivial dynamics’ (random walk). Therefore, whenever fitting model (I) results in an estimated α that is nearly zero, then the best fitting model of type (I) is in fact a (UQ) model. In which case we have evidence to prefer a (UQ) modeling approach instead, which will correspond to non-trivial dynamics. Conversely, if fitting model (UQ) yields an estimated α near 1, it may be preferable to use model (I) instead.

6. PROOFS

Proof of Lemma 3.4. The proof is directly analogous to that of theorem 2 in Wu and Shao (2004) and theorem 1 in Zhu and Müller (2021). ■

Proof of Lemma 3.7. We prove the theorem in the following four steps:

1. Given a function $f \in L^2$, and a random function ϵ such that $\mathbb{E}(\epsilon) = \text{id}$, we can show that

$$\arg \min_h \mathbb{E}_\epsilon \|h - \epsilon \circ f\|_2^2 = f.$$

To do so, we can apply Fubini’s theorem and rewrite the expression as follows:

$$\int \int |h(x) - \epsilon(f(x))|^2 dx d\epsilon = \int \int |h(x) - \epsilon(f(x))|^2 d\epsilon dx.$$

Since $\mathbb{E}_\epsilon[\epsilon(f(x))] = f(x)$ for any x , the minimizer of the inner integral on the left-hand side is $h(x) = \mathbb{E}[\epsilon(f(x))] = f(x)$.

2. We will now demonstrate that for any fixed T_i and T_{i-1} , as well as for all α , the following inequality holds:

$$\mathbb{E}_\epsilon[g_\alpha(T_{i-1}, T_i, S_\alpha)] \leq \mathbb{E}_\epsilon[g_\alpha(T_{i-1}, T_i, S_\alpha)].$$

Let us define $f(\alpha, T) = S_\alpha \circ [\alpha T]$. Note that for all indices i , we have

$$\begin{aligned} g_\alpha(T_i, T_{i-1}, S_\alpha) &= \left\| S_\alpha \circ [\alpha T_{i-1}] - T_{\epsilon_i} \circ S_\alpha \circ [\alpha T_{i-1}] \right\|_2^2 \\ &= \left\| f(\alpha, T_{i-1}) - T_{\epsilon_i} \circ f(\alpha, T_{i-1}) \right\|_2^2. \end{aligned} \quad (14)$$

Using the result from part 1 and the equation (14), we can conclude that if there exists an α such that $f(\alpha, T_{i-1})$ minimizes the expression $\mathbb{E}_\epsilon[g_\alpha(T_{i-1}, T_i, S_\alpha)]$ in equation (14), then we must have $f(\alpha, T_{i-1}) = f(\alpha, T_{i-1})$. Therefore, we obtain the desired inequality.

3. We now aim to prove that for any α , we have

$$\mathbb{E}[g_\alpha(T_{i-1}, T_i, S_\alpha)] \leq \mathbb{E}[g_\alpha(T_{i-1}, T_i, S_\alpha)].$$

We start by denoting by π the marginal distribution of T_i , and Q the marginal distribution of the pair (T_{i-1}, T_i) . Then, we can express the expectation of $g_\alpha(T_{i-1}, T_i, S_\alpha)$ as follows:

$$\mathbb{E}_Q[g_\alpha(T_{i-1}, T_i, S_\alpha)] = \mathbb{E}_\pi[\mathbb{E}_{\epsilon_{g_\alpha}}(T_{i-1}, T_i, S_\alpha) | T_{i-1}].$$

By using part 2 of the proof, we know that α is a minimizer for the inner expectation of the right-hand side, i.e.,

$$\mathbb{E}_\epsilon \{g_\alpha(T_{i-1}, T_i, S_\alpha) | T_{i-1}\} \leq \mathbb{E}_\epsilon \{g_\alpha(T_{i-1}, T_i, S_\alpha) | T_{i-1}\},$$

and this for all T_{i-1} . Therefore, taking the expectation over T_{i-1} , we get

$$\mathbb{E}_Q[g_\alpha(T_{i-1}, T_i, S_\alpha)] \leq \mathbb{E}_Q[g_\alpha(T_{i-1}, T_i, S_\alpha)].$$

4. Finally we can conclude that α is the unique minimizer of $M(\alpha)$. Suppose there exists an α such that $\mathbb{E}[g_\alpha(T_{i-1}, T_i, S_\alpha)] = \mathbb{E}[g_\alpha(T_{i-1}, T_i, S_\alpha)]$. Using parts 2 and 3, we can deduce that for each fixed T_i, T_{i-1} , $\mathbb{E}_\epsilon[g_\alpha(T_{i-1}, T_i, S_\alpha)] = \mathbb{E}_\epsilon[g_\alpha(T_{i-1}, T_i, S_\alpha)]$. Then using equation (14) we can conclude that, for all indices j ,

$$S_\alpha \circ [\alpha T_j] = S_\alpha \circ [\alpha T_j].$$

If $S_\alpha \circ \alpha T_j = S_\alpha \circ [\alpha T_j]$ for all j , we can deduce $S_\alpha = S_\alpha \circ [\alpha T_j] \circ [\alpha T_j]^{-1}$ for all j . However, note that while S_α is deterministic, the right-hand side is deterministic (and not random) if and only if $\alpha = \alpha$. This is because if $\alpha \neq \alpha$, then the right-hand side depends on T_j , which is a random variable. ■

Lemma 6.1. A sequence $\{T_n\}_{i=-\infty}^\infty$ that satisfies the geometric moment contracting condition (3.3) for $\eta \geq 2$, also satisfies the conditions (1.1), (1.2), (2.1), and (2.2) of Horváth *et al.* (2013). Namely, assume

$$T_n = f(\epsilon_n, \epsilon_{n-1}, \dots),$$

where $\{\epsilon'_i\}$ is an independent copy of $\{\epsilon_i\}$ defined in the same probability space. Then, letting

$$T'_{n,m} = f(\epsilon_n, \epsilon_{n-1}, \dots, \epsilon_{n-m+1}, \epsilon'_{n-m}, \dots), \quad (15)$$

for any $0 < \delta < 1$ we have

$$\sum_{m=1}^{\infty} (\mathbb{E} \|T_n - T'_{n,m}\|_2^2)^{1/2} < \infty. \quad (16)$$

Proof of Lemma 6.1. Let $T_{n-m} = f(\epsilon_{n-m}, \epsilon_{n-m-1}, \dots)$ and $T'_{n-m} = f(\epsilon'_{n-m}, \epsilon'_{n-m-1}, \dots)$. Thus we can write $T_n = \Phi_{n,m}(T_{n-m})$ and $T'_{n,m} = \Phi_{n,m}(T'_{n-m})$.

$$\begin{aligned} \sum_{m=1}^{\infty} (\mathbb{E} \|T_n - T'_{n,m}\|_2^2)^{1/2} &= \sum_{m=1}^{\infty} (\mathbb{E} \|\Phi_{n,m}(T_{n-m}) - \Phi_{n,m}(T'_{n-m})\|_2^2)^{1/2} \\ &\leq \sum_{m=1}^{\infty} (\mathbb{E} \|\Phi_{n,m}(T_{n-m}) - \Phi_{n,m}(Q_0)\|_2^2)^{1/2} \\ &\quad + (\mathbb{E} \|\Phi_{n,m}(T'_{n-m}) - \Phi_{n,m}(Q_0)\|_2^2)^{1/2} \\ (\text{Lyapunov's inequality}) \quad &\leq \sum_{m=1}^{\infty} (\mathbb{E} \|\Phi_{n,m}(T_{n-m}) - \Phi_{n,m}(Q_0)\|_2^\eta)^{1/\eta} \\ &\quad + (\mathbb{E} \|\Phi_{n,m}(T'_{n-m}) - \Phi_{n,m}(Q_0)\|_2^\eta)^{1/\eta} \end{aligned}$$

$$\begin{aligned}
 (\text{Assumption 3.3}) \quad & \leq \sum_{m=1}^{\infty} C r^{m/\eta} (\|T_{n-m} - Q_0\|_2 \vee \|T'_{n-m} - Q_0\|_2) \\
 & < \infty.
 \end{aligned} \tag{17}$$

Lemma 6.2. For any $T, S \in \mathcal{T}$ we have $\|T^{-1} - S^{-1}\|_2 \lesssim \sqrt{\|T - S\|_2}$. Moreover, let $\mathcal{T}_{l,u} = \{T \in \mathcal{T} : 0 < L_l \leq T' \leq L_u < \infty\}$. For any $T, S \in \mathcal{T}_{l,u} \subset \mathcal{T}$ we have $\|T^{-1} - S^{-1}\|_2 \lesssim \|T - S\|_2$. In summary, there exists $b \in [\frac{1}{2}, 1]$ such that $\|T^{-1} - S^{-1}\|_2 \lesssim \|T - S\|_2^b$ for any $T, S \in \mathcal{T}$.

Proof. Let $T, S \in \mathcal{T}$. Then, for some constant C' , we have: $\|T^{-1} - S^{-1}\|_2 \leq C' \|T^{-1} - S^{-1}\|_1$, because the functions are bounded. Moreover, $\|T^{-1} - S^{-1}\|_1 = \|T - S\|_1$. And, finally, by applying the Cauchy-Schwarz inequality, we get $\|T - S\|_1 \leq C\sqrt{\|T - S\|_2}$, where C is a constant. Therefore, we conclude $\|T^{-1} - S^{-1}\|_2 \leq C\sqrt{\|T - S\|_2}$.

When $T, S \in \mathcal{T}_{l,u}$ we can write

$$\begin{aligned}
 \|T^{-1} - S^{-1}\|_2^2 &= \int_0^1 |T^{-1}(x) - S^{-1}(x)|^2 dx \\
 &= \int_0^1 |T^{-1} \circ S(y) - y|^2 S'(y) dy \quad (S^{-1}(x) = y) \\
 &\leq L_u \int |T^{-1} \circ S(y) - y|^2 dy \\
 &\leq L_u \int |z - S^{-1} \circ T(z)|^2 \frac{1}{S'(S^{-1} \circ T(z))} T'(z) dz \quad (T^{-1} \circ S(y) = z) \\
 &\leq L_u \frac{L_u}{L_l} \int |z - S^{-1} \circ T(z)|^2 dz \\
 &\leq L_u \frac{L_u}{L_l} \frac{1}{L_l} \int |S(z) - T(z)|^2 dz \quad (\forall x, y \quad |x - y| \leq \frac{1}{L_l} |S(x) - S(y)|) \\
 &\leq \frac{L_u^2}{L_l^2} \|S - T\|_2^2.
 \end{aligned} \tag{18}$$

Lemma 6.3. There exists a constant $\frac{1}{2} \leq b \leq 1$ such that the following inequalities hold:

$$\|S_{\alpha_1} - S_{\alpha_2}\|_2 \lesssim |\alpha_1 - \alpha_2|^b,$$

and

$$\|S_{N,\alpha_1} - S_{N,\alpha_2}\|_2 \lesssim |\alpha_1 - \alpha_2|^b,$$

and

$$g_{\alpha_1}(T_{i-1}, T_i, S_{N,\alpha_1}) - g_{\alpha_2}(T_{i-1}, T_i, S_{N,\alpha_2}) \leq C(T_i) |\alpha_1 - \alpha_2|^b,$$

where $\frac{1}{n} \sum_i \mathbb{E}[C(T_i)] = O(1)$.

Define $\mathcal{T}_{l,u} = \{T \in \mathcal{T} : 0 < L_l \leq T' \leq L_u < \infty\}$. If $\{T_i\} \subset \mathcal{T}_{l,u}$, then $b = 1$ in the above inequalities.

Proof. To begin with, it should be noted that given any two real numbers $\alpha_1, \alpha_2 \in (-1, 1)$ with the same sign, and for any given map T , we have the following inequality:

$$\|[\alpha_1 T] - [\alpha_2 T]\|_2 \leq C |\alpha_1 - \alpha_2|,$$

where C is a constant. In fact, it suffices to consider the definition of $[\alpha T]$ for the cases when $\alpha \geq 0$ and $\alpha < 0$ separately. Using Lemma 6.2 we can write that for some $\frac{1}{2} \leq b \leq 1$,

$$\begin{aligned} \|S_{\alpha_1} - S_{\alpha_2}\|_2 &= \|\mathbb{E}[T_j \circ [\alpha_1 T_{j-1}]^{-1}] - \mathbb{E}[T_j \circ [\alpha_2 T_{j-1}]^{-1}]\|_2 \\ &\leq LC' |\alpha_1 - \alpha_2|^b, \end{aligned} \quad (19)$$

where L is the common Lipschitz constant for all T_j . Similarly

$$\|S_{N,\alpha_1} - S_{N,\alpha_2}\|_2 \leq L |\alpha_1 - \alpha_2|^b,$$

We now proceed to show that both $S_{N,\alpha}$ and S_α are Lipschitz functions of x . To do this, we observe that the inverse of a Lipschitz function is Lipschitz, and also the composition of two Lipschitz functions is Lipschitz. Since all T_j are Lipschitz and $S_{N,\alpha}$ and S_α are defined as compositions, they are also Lipschitz with respect to x .

We will now show that g is Lipschitz function of α :

$$\begin{aligned} g_{\alpha_1}(T_{i-1}, T_i, S_{N,\alpha_1}) - g_{\alpha_2}(T_{i-1}, T_i, S_{N,\alpha_2}) &= \|S_{N,\alpha_1} \circ [\alpha_1 T_i] - T_{i+1}\|^2 - \|S_{N,\alpha_2} \circ [\alpha_2 T_i] - T_{i+1}\|^2 \\ &\lesssim \|S_{N,\alpha_1} \circ [\alpha_1 T_i] - S_{N,\alpha_2} \circ [\alpha_2 T_i]\|_2^2 \\ &\lesssim \|S_{N,\alpha_1} \circ [\alpha_1 T_i] - S_{N,\alpha_1} \circ [\alpha_2 T_i]\|_2^2 \\ &\quad + \|S_{N,\alpha_1} \circ [\alpha_2 T_i] - S_{N,\alpha_2} \circ [\alpha_2 T_i]\|_2^2 \\ &\leq D(T_i) |\alpha_1 - \alpha_2| + C(T_i) |\alpha_1 - \alpha_2|^b \\ &\lesssim C(T_i) |\alpha_1 - \alpha_2|^b, \end{aligned} \quad (20)$$

where $D(T_i)$ and $C(T_i)$ are constants that depend on T_i and $\sum_i \mathbb{E}[C(T_i)]/n = O(1)$. ■

The following statement is virtually obvious, but is used multiple times in the proofs below and so is most easily quoted directly:

Lemma 6.4. Let $\{X_i\}$ be a sequence of random variables, and suppose that $W_n := \sqrt{n}(\frac{1}{n} \sum_{i=1}^n X_i - \mu) \xrightarrow{d} W$ for some (almost surely finite) random variable W . Then, $\frac{1}{n} \sum_{i=1}^n X_i$ converges in probability to μ .

Proof. By Slutsky's Theorem, we get $n^{-1/2} W_n \xrightarrow{d} 0$, which also implies convergence in probability to zero. ■

Proof of Lemma 3.8. We start by using Lemma 6.1 to conclude that the series $\{T_i - \mathbb{E}T_i\}_{i=-\infty}^\infty$ satisfies the assumptions (1.1), (1.2), (2.1), and (2.2) of Horváth *et al.* (2013). From this, we can argue that the series $\{T_j \circ [\alpha T_{j-1}]^{-1} - \mathbb{E}T_j \circ [\alpha T_{j-1}]^{-1}\}_{i=-\infty}^\infty$ also satisfies those assumptions and therefore we obtain the following central limit theorem for $S_{N,\alpha}$: for any α , there is a Gaussian process Γ_α such that

$$\sqrt{N}(S_{N,\alpha} - S_\alpha) \xrightarrow{d} \Gamma_\alpha, \quad \text{in } L^2.$$

Using the central limit theorem and Lemma 6.4, we can infer the convergence in probability of $S_{N,\alpha}$ to S_α for any α (in L^2). Since both S_α and $S_{N,\alpha}$ are globally Lipschitz with respect to α , in the sense of Lemma 6.3, we can use Corollary 3.1 of Newey (1991) to obtain uniform convergence in probability:

$$\sup_\alpha \|S_{N,\alpha} - S_\alpha\|_2 \rightarrow 0 \quad \text{in probability.} \quad \blacksquare$$

Proposition 6.5. Assume that there exists $\delta > 0$ such that $1/\delta \leq |T'_\epsilon(x)| \leq \delta$ almost surely and $1/\delta \leq |S'(x)| \leq \delta$, for all $x \in \Omega$. Then whenever $\alpha < 0$, the map T_i is almost surely contained within $\mathcal{T}_{l,u} = \{T \in \mathcal{T} : 0 < l \leq T' \leq u < \infty\}$, for any choice of (l, u) in the set

$$\mathcal{L} = \left\{ 0 < x < y < \infty : x \leq \frac{1}{\delta^2} \left(1 + \alpha - \frac{\alpha}{y} \right) \leq \delta^2 \left(1 + \alpha - \frac{\alpha}{x} \right) \leq y \right\}.$$

In particular, $\mathcal{L}$ is non-empty for all $-1 \leq \alpha < 0$ and $\delta > 0$.

Proof of Lemma 6.5. Note that whenever $\alpha < 0$ we have

$$T'_i(x) = T'_{\epsilon_i}(S \circ [\alpha T_{i-1}](x)S'([\alpha T_{i-1}](x)) \left(1 + \alpha - \frac{\alpha}{T'_{i-1}(T_{i-1}^{-1}(x))} \right).$$

Therefore if $T_{i-1} \in \mathcal{T}_{l,u}$ then we can conclude $T'_i(x) \leq \frac{1}{\delta^2}(1 + \alpha - \frac{\alpha}{L_l})$ and $\delta^2(1 + \alpha - \frac{\alpha}{L_u}) \leq T'_i(x)$ and hence $T_i \in \mathcal{T}_{l,u}$ for any choice of (l, u) in the set

$$\mathcal{L} = \left\{ 0 < x < y < \infty : x \leq \frac{1}{\delta^2} \left(1 + \alpha - \frac{\alpha}{y} \right) \leq \delta^2 \left(1 + \alpha - \frac{\alpha}{x} \right) \leq y \right\}.$$

(It's straightforward to see that the second inequality always hold true if $-1 \leq \alpha < 0$, and $x < y$.) ■

6.1. Overview of Wu and Shao (2004)

In their work, Wu and Shao (2004) investigated the properties of nonlinear time series expressed in terms of iterated random functions and established a central limit theorem for additive functionals of such systems. The construction involves a sequence of functions of the form $X_n(x) = F_{\theta_n} \circ F_{\theta_{n-1}} \circ \cdots \circ F_{\theta_1}(x)$. The authors assume that X_n satisfies a geometric moment condition, which requires the existence of $\beta > 0$, $C = C(\alpha) > 0$, and $r = r(\alpha) \in (0, 1)$ such that, for all $n \in \mathbb{N}$,

$$\mathbb{E}\{\rho(X_n(X'_0), X_n(X_0))^\beta\} \leq Cr^n. \quad (21)$$

In addition, they define the l -dimensional vector $Y_i = (X_{i-l+1}, X_{i-l+2}, \dots, X_i)$ and for any $\delta > 0$, they introduce the functional $\Delta_g(\delta)$ as

$$\Delta_g(\delta) = \sup \left\{ \left\| [g(Y) - g(Y_1)]1_{\rho(Y, Y_1) \leq \delta} \right\| : Y, Y_1 \text{ are identically distributed} \right\},$$

Where $\rho(., .)$ is the product metric and is defined as

$$\rho(z, z') = \sqrt{\sum_{i=1}^l \rho(z_i, z'_i)^2} \quad \text{for } z = (z_1, \dots, z_l), z' = (z'_1, \dots, z'_l).$$

Finally, the functional $S_{n,l}(g) = \sum_{i=1}^n g(X_{i-l+1}, X_{i-l+2}, \dots, X_i)$ is defined. The authors establish the following central limit theorem for this functional:

Theorem 6.6. (Wu and Shao (2004, theorem 3)). Assume that (21) holds, that $X_1 \sim \pi$, $E\{g(Y_1)\} = 0$, and $E\{|g(Y_1)|^p\} < \infty$ for some $p > 2$, and that

$$\int_0^1 \frac{\Delta_g(t)}{t} < \infty. \quad (22)$$

Then there exists a $\sigma_g \geq 0$ such that, for π -almost x , $\{S_{[nu],l}(g)/\sqrt{n}, 0 \leq u \leq 1\}$ conditional on $X_0 = x$, converges to $\sigma_g B$, where B is a standard Brownian motion.

A function that satisfies (22) is referred to as *stochastic Dini continuous*. Using Theorem 6.6 to derive a central limit theorem for M_N poses a problem: Theorem 6.6 uses fixed-length subsequences of the time series, i.e., $(X_{i-l+1}, X_{i-l+2}, \dots, X_i)$, as arguments for the function g , however, the arguments of the function g that appears in the expression of M_N in 9, include not only (T_{i-1}, T_i) , but also $S_{N,\alpha}$, thus making it dependent on the entire time series. Therefore, Theorem 6.6 cannot be applied directly, and a modified version is required. We present a modified version of Theorem 6.6 that is specifically tailored for functions of finite dimensional random variables, followed by another modification that is suitable for functionals of infinite dimensional variables.

Corollary 6.7. (Modified version of Wu and Shao (2004, theorem 3) for finite dimensional arguments). Suppose $\bar{Z}_n$ is a measurable function of $(X_1, X_2, \dots, X_n)$ such that $\bar{Z}_n$ converges in probability to some constant μ . Let $Y_i = (X_{i-l+1}, X_{i-l+2}, \dots, X_i)$, and assume that $g(Y_i, \mu)$ is differentiable with respect to its second argument and that both $g(Y_i, \mu)$ and the derivative of $g(Y_i, \mu)$ with respect to its second argument satisfy the conditions of Theorem 6.6. Then there exists $\sigma_g \geq 0$ such that

$$\frac{S_n}{\sqrt{n}} \rightarrow N(0, \sigma_g^2),$$

where $S_n = \sum_{i=1}^n g(Y_i, \bar{Z}_n)$.

Proof. By Taylor expansion, we write

$$g(Y_i, \bar{Z}_n) = g(Y_i, \mu) + g^{(0,1)}(Y_i, \mu)(\bar{Z}_n - \mu) + o_{\mathbb{P}}(1).$$

Since $g^{(0,1)}(Y_i, \mu)$ is only a function of Y_i and of a constant μ , by Theorem 6.6 we have

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n [g^{(0,1)}(Y_i, \mu) - \mathbb{E}g^{(0,1)}(Y, \mu)] \rightarrow N(0, \sigma_{g'}^2),$$

where $Y \stackrel{D}{\sim} Y_i$. This implies that if $N \stackrel{D}{\sim} N(0, \sigma_{g'}^2)$, we then have

$$\begin{aligned} \frac{1}{\sqrt{n}} S_n &= \left[\frac{1}{\sqrt{n}} \sum_{i=1}^n g(Y_i, \mu) \right] + (N + \sqrt{n} \mathbb{E}g^{(0,1)}(Y, \mu))(\bar{Z}_n - \mu) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(Y_i, \mu) + \mathbb{E}g^{(0,1)}(Y, \mu)(Z_i - \mu)] + N(\bar{Z}_n - \mu). \end{aligned} \quad (23)$$

Since $N(\bar{Z}_n - \mu) = o_{\mathbb{P}}(1)$, applying Theorem 6.6 we get

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n [g(Y_i, \mu) + \mathbb{E}g^{(0,1)}(Y, \mu)(Z_i - \mu)] \rightarrow N(0, \sigma_g^2).$$

■

Corollary 6.8. (Modified version of Wu and Shao (2004, theorem 3) for infinite dimensional arguments). Suppose $\bar{Z}_n$ is a measurable function of $(X_1, X_2, \dots, X_n)$ such that $\bar{Z}_n$ converges in probability to some constant μ . Let $Y_i = (X_{i-l+1}, X_{i-l+2}, \dots, X_i)$, and assume that $g(Y_i, \mu)$ is Fréchet differentiable with respect to its second argument, and that both $g(Y, \mu)$ and the Fréchet derivative of g with respect to its second argument satisfy the conditions of Theorem 6.6. Then there exists $\sigma_g \geq 0$ such that

$$\frac{S_n}{\sqrt{n}} \rightarrow N(0, \sigma_g^2),$$

where $S_n = \sum_{i=1}^n g(Y_i, \bar{Z}_n)$.

Remark 6.9. The proof of this Corollary can be understood by following the same steps as in the proof of Corollary 6.7, without the added technical complexities that arise when dealing with the Fréchet derivative.

Proof of Corollary 6.8. Let $D_g(Y_i, u, v)$ denote the Fréchet derivative of g with respect to its second argument at u in the direction v . Assume $\bar{Z}_n = \mu + v_n$, and apply the Taylor formula for the Fréchet derivative (Kurdila and Zabrankin, 2006) to get

$$g(Y_i, \bar{Z}_n) = g(Y_i, \mu) + D_g(Y_i, \mu, v_n) + R(Y_i, \mu, v_n),$$

where

$$\lim_{\|v_n\| \rightarrow 0} \frac{|R(Y_i, \mu, v_n)|}{\|v_n\|} = 0.$$

Note that we can identify the Fréchet derivative with a bounded linear operator as

$$D_g(Y_i, \mu, v_n) = \langle D_g(Y_i, \mu), v_n \rangle.$$

Furthermore, as the Fréchet derivative is also stochastic Dini continuous, we can apply Theorem 6.6 to obtain

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n [\langle D_g(Y_i, \mu), v_n \rangle - \mathbb{E} \langle D_g(Y, \mu), v_n \rangle] \rightarrow N(0, \sigma_g^2),$$

where $Y \stackrel{D}{\sim} Y_i$.

This implies that if $N \stackrel{D}{\sim} N(0, \sigma_g^2)$, using the fact that the mapping $D_g(Y_i, \mu, \cdot)$ is linear, we get:

$$\begin{aligned} \frac{S_n}{\sqrt{n}} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(Y_i, \mu) + D_g(Y_i, \mu, \bar{Z}_n - \mu)] \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(Y_i, \mu) + \langle D_g(Y_i, \mu), \bar{Z}_n - \mu \rangle] \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(Y_i, \mu)] + \langle N \times \text{id} + \sqrt{n} \mathbb{E} D_g(Y, \mu), \bar{Z}_n - \mu \rangle \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(Y_i, \mu) + \langle \mathbb{E} D_g(Y, \mu), Z_i - \mu \rangle] + N \langle \text{id}, \bar{Z}_n - \mu \rangle. \end{aligned} \quad (24)$$

Since $N\langle \text{id}, \bar{Z}_n - \mu \rangle = o_p(1)$, and $\mathbb{E}\langle \mathbb{E}D_g(Y, \mu), Z_i - \mu \rangle = 0$, we can apply Theorem 6.6 and conclude $\frac{S_n}{\sqrt{n}} \rightarrow N(0, \sigma^2)$ for some σ . ■

Lemma 6.10. The function $g_\alpha(T_{i-1}, T_i, S) = \|S \circ [\alpha T_{i-1}] - T_i\|_2^2$ is Fréchet differentiable with respect to S and satisfies the Taylor formula

$$g_\alpha(T_{i-1}, T_i, S + v) = g_\alpha(T_{i-1}, T_i, S) + D_g(T_{i-1}, T_i, S, v) + R(T_{i-1}, T_i, S, v),$$

where $D_g(T_{i-1}, T_i, S, v)$ is the Fréchet derivative of g_α with respect to S in the direction v , and

$$\lim_{\|v\| \rightarrow 0} \frac{|R(T_{i-1}, T_i, S, v)|}{\|v\|} = 0.$$

Furthermore, the mapping $D_g(T_{i-1}, T_i, \mu, \cdot)$ is both linear and bounded.

Proof. To begin, we show that $g_\alpha(T_{i-1}, T_i, S)$ is Gateaux differentiable.

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \frac{g_\alpha(T_{i-1}, T_i, S + \epsilon v) - g_\alpha(T_{i-1}, T_i, S)}{\epsilon} &= \lim_{\epsilon \rightarrow 0} \frac{\|(S + \epsilon v) \circ [\alpha T_{i-1}] - T_i\|_2^2 - \|S \circ [\alpha T_{i-1}] - T_i\|_2^2}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{\epsilon^2 \|v \circ [\alpha T_{i-1}]\|_2^2 + \epsilon \langle v \circ \alpha T_{i-1}, S \circ [\alpha T_{i-1}] - T_i \rangle}{\epsilon} \\ &= \langle v \circ [\alpha T_{i-1}], S \circ [\alpha T_{i-1}] - T_i \rangle \\ &= D_g((T_i, T_{i-1}), S, v). \end{aligned} \quad (25)$$

As the above expression is linear and bounded with respect to v , it serves as the Gateaux differential. As $D_g(T_{i-1}, T_i, S)$ is Gateaux differentiable for every T and the mapping $T \rightarrow D_g(T_{i-1}, T_i, S)$ is continuous, Corollary 4.1.1. of Kurdila and Zabrankin (2006) guarantees that D_g is also the Fréchet derivative. ■

Lemma 6.11. The stochastic Dini continuity condition (22) is satisfied by the function g_α .

Proof. We want to show $\int_0^1 \frac{\Delta_g(t)}{t} < \infty$, where

$$\begin{aligned} \Delta_g(\delta) &= \sup \left\{ \left\| [g_\alpha(T_{i-1}, T_i, S) - g_\alpha(T'_{i-1}, T'_i, S)] 1_{\rho((T_{i-1}, T_i), (T'_{i-1}, T'_i)) \leq \delta} \right\| \right. \\ &\quad \left. : T_i, T'_i \text{ are identically distributed} \right\}. \end{aligned} \quad (26)$$

$$\text{and } \rho((T_1, T_2), (T'_1, T'_2)) = \sqrt{\|T_1 - T'_1\|_2^2 + \|T_2 - T'_2\|_2^2}.$$

First, note that $g_\alpha(T_{i-1}, T_i, S) = \|S \circ [\alpha T_{i-1}] - T_i\|_2^2$ and $g_\alpha(T'_{i-1}, T'_i, S) = \|S \circ [\alpha T'_{i-1}] - T'_i\|_2^2$. When $\alpha \geq 0$, we have $\|[\alpha T_{i-1}] - [\alpha T'_{i-1}]\|_2 \leq \alpha \|T_{i-1} - T'_{i-1}\|_2$. When $\alpha < 0$, we can use Lemma 6.2 to conclude that $\|[\alpha T_{i-1}] - [\alpha T'_{i-1}]\|_2 \leq \alpha \|T_{i-1} - T'_{i-1}\|_2^b$ for some $b \geq \frac{1}{2}$. As S is Lipschitz, we can deduce that $\Delta_g(t) \leq C\alpha t^b$, for some $b > 0$. Therefore the integral is finite. ■

Proof of Theorem 3.9. From Lemma 3.8, we see that $S_{N,\alpha}$ converges in probability to S_α and we also obtained a central limit theorem for $S_{N,\alpha}$. Then Lemma 6.10 and 6.11 show that g_α is Fréchet differentiable and stochastically Dini continuous, which are sufficient conditions for Corollary 6.8 to be applicable, and yield a central limit theorem for $M_N(\alpha) = \frac{1}{N} \sum_{i=1}^N g_\alpha(T_{i-1}, T_i, S_{N,\alpha})$:

$$\sqrt{N}[M_N(\alpha) - M(\alpha)] \rightarrow N(0, \sigma_\alpha^2).$$

Thus for any α , $M_N(\alpha)$ converges in probability to $M(\alpha)$. By applying corollary 3.1 from Newey (1991) and utilizing Lemma 6.3, which establishes that g_α satisfies Lipschitz continuity with respect to α , we can achieve uniform convergence in probability of M_N to M with respect to α :

$$\sup_\alpha |M_N(\alpha) - M(\alpha)| \rightarrow 0 \quad \text{in probability.}$$

Proof of Theorem 3.10 (Consistency). Lemma 3.9 implies that M_N converges uniformly in probability to M with respect to α , and Lemma 3.7 shows that α is the unique minimizer of M . By applying Van Der Vaart and Wellner (1996, theorem 3.2.3), we can conclude that the estimator $\hat{\alpha}_N = \arg \min_\alpha M_N(\alpha)$ converges to the true parameter $\arg \min_\alpha M(\alpha) = \alpha$. ■

We will now employ M-estimation theory to establish the convergence rate of our estimator. To do so, we recall the following theorem, which is taken from Van Der Vaart and Wellner (1996).

Theorem 6.12. (Van Der Vaart and Wellner (1996), Theorem 3.2.5.). Let M_N be a stochastic process indexed by a metric space Θ , and let M be a deterministic function, such that for every θ in a neighborhood of θ_0 ,

$$M(\theta) - M(\theta_0) \gtrsim d^2(\theta, \theta_0).$$

Suppose that, for every N and sufficiently small δ ,

$$\mathbb{E}^* \sup_{d^2(\theta, \theta_0) < \delta} \sqrt{N} |(M_N - M)(\theta) - (M_N - M)(\theta_0)| \lesssim \phi_N(\delta),$$

for functions ϕ_N such that $\delta \rightarrow \phi_N(\delta)/\delta^\alpha$ is decreasing for some $\alpha < 2$ (not depending on N). Let

$$r_N^2 \phi_N \left(\frac{1}{r_N} \right) \leq \sqrt{N}, \quad \text{for every } N.$$

If the sequence $\hat{\theta}_N$ satisfies $M_N(\hat{\theta}_N) \leq M_N(\theta_0) + o_{\mathbb{P}}(r_N^{-2})$, and converges in outer probability to θ_0 , then $r_N d(\hat{\theta}_N, \theta_0) = O_{\mathbb{P}}^*(1)$. If the displayed conditions are valid for every θ and δ , then the condition that $\hat{\theta}_N$ is consistent is unnecessary.

Lemma 6.13. Let $\mathcal{T}_{l,u} = \{T \in \mathcal{T} : 0 < L_l \leq T' \leq L_u < \infty\}$ and suppose $\{T_i\} \subset \mathcal{T}_{l,u}$. Then for any α ,

$$\mathbb{E}|M'_N(\alpha) - M'(\alpha)| \lesssim \frac{1}{\sqrt{N}}.$$

Proof. Note that

$$M'_N(\alpha) = \frac{1}{N} \sum_{j=1}^N \frac{\partial g_\alpha(T_{j-1}, T_j, S_{N,\alpha})}{\partial \alpha},$$

and

$$\begin{aligned} \frac{\partial g_\alpha(T_{j-1}, T_j, S_{N,\alpha})}{\partial \alpha} &= \frac{\partial \|S_{N,\alpha} \circ [\alpha T_{j-1}] - T_j\|_2^2}{\partial \alpha} \\ &= \int 2|S_{N,\alpha} \circ [\alpha T_{j-1}](x) - T_j(x)| \times \frac{\partial}{\partial \alpha} S_{N,\alpha} \circ [\alpha T_j](x) dx. \end{aligned}$$

The expression $|S_{N,\alpha} \circ [\alpha T_{j-1}](x) - T_j(x)|$ can be uniformly bounded. In what follows we will explicitly calculate $\frac{\partial}{\partial \alpha} S_{N,\alpha} \circ [\alpha T_j](x)$ for a fixed j . The calculation is tedious but elementary. To calculate the derivative we use the following fact: if $f(\alpha, x) = C(\alpha, y(x, \alpha))$, then

$$\frac{\partial f}{\partial \alpha} = \frac{\partial C(\alpha, y(x, \alpha'))}{\partial \alpha} \Big|_{\alpha'=\alpha} + \frac{\partial C(\alpha, y)}{\partial y} \times \frac{\partial y(x, \alpha)}{\partial \alpha}.$$

Using the above equation we can write:

$$\begin{aligned} \frac{\partial}{\partial \alpha} S_{N,\alpha} \circ [\alpha T_j](x) &= \frac{\partial}{\partial \alpha} S_{N,\alpha}([\alpha' T_j](x)) \Big|_{\alpha'=\alpha} + \frac{\partial S_{N,\alpha}([\alpha T_j](x))}{\partial([\alpha T_j](x))} \times \frac{\partial[\alpha T_j](x)}{\partial \alpha} \\ &= \frac{\partial S_{N,\alpha}(y)}{\partial \alpha} \Big|_{y=[\alpha T_j](x)} + \frac{\partial S_{N,\alpha}(y)}{\partial y} \Big|_{y=[\alpha T_j](x)} \times \frac{\partial[\alpha T_j](x)}{\partial \alpha}. \end{aligned} \quad (27)$$

First, we derive the first term on the LHS of (27):

$$\frac{\partial S_{N,\alpha}(y)}{\partial \alpha} = \sum_{i=1}^N \frac{\partial}{\partial \alpha} T_i \circ [\alpha T_{i-1}]^{-1}(y).$$

If we consider one of the terms in this summation we have

$$\begin{aligned} \frac{\partial}{\partial \alpha} T_i \circ [\alpha T_{i-1}]^{-1}(y) &= \frac{\partial T_i([\alpha T_{i-1}]^{-1}(y))}{\partial[\alpha T_{i-1}]^{-1}(y)} \times \frac{\partial[\alpha T_{i-1}]^{-1}(y)}{\partial \alpha} \\ &= T'_i(z_i) \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)} \times \frac{\partial}{\partial \alpha} [\alpha T_{i-1}]^{-1}(y). \end{aligned} \quad (28)$$

Now to calculate $\frac{\partial}{\partial \alpha} [\alpha T_{i-1}]^{-1}(y)$ note that:

$$\begin{aligned} 0 &= \frac{\partial}{\partial \alpha} y = \frac{\partial}{\partial \alpha} [\alpha T_{i-1}]([\alpha T_{i-1}]^{-1}(y)) \\ &= \frac{\partial}{\partial \alpha} [\alpha T_{i-1}]([\alpha' T_{i-1}]^{-1}(y)) \Big|_{\alpha'=\alpha} + \frac{\partial[\alpha T_{i-1}]([\alpha T_{i-1}]^{-1}(y))}{\partial[\alpha T_{i-1}]^{-1}(y)} \times \frac{\partial[\alpha T_{i-1}]^{-1}(y)}{\partial \alpha} \\ &= \frac{\partial}{\partial \alpha} [\alpha T_{i-1}](z_i) \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)} + \frac{\partial[\alpha T_{i-1}](z_i)}{\partial z_i} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)} \times \frac{\partial[\alpha T_{i-1}]^{-1}(y)}{\partial \alpha}. \end{aligned} \quad (29)$$

Thus

$$\begin{aligned} \frac{\partial[\alpha T_{i-1}]^{-1}(y)}{\partial \alpha} &= (-1) \times \frac{\partial}{\partial \alpha} [\alpha T_{i-1}](z_i) \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)} \times \frac{1}{\frac{\partial[\alpha T_{i-1}](z_i)}{\partial z_i} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}} \\ &= \begin{cases} (z_i - T_{i-1}(z_i)) \times \frac{1}{\alpha(T'_{i-1}(z_i)-1)+1} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } 0 < \alpha \leq 1 \\ (T_{i-1}^{-1}(z_i) - z_i) \times \frac{1}{\alpha(1-(T_{i-1}^{-1})'(z_i))+1} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } -1 \leq \alpha < 0, \end{cases} \end{aligned} \quad (30)$$

And we can conclude that

$$\frac{\partial}{\partial \alpha} T_i \circ [\alpha T_{i-1}]^{-1}(y) = T'_i(z_i) \times \begin{cases} (z_i - T_{i-1}(z_i)) \times \frac{1}{\alpha(T'_{i-1}(z_i)-1)+1} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } 0 < \alpha \leq 1 \\ (T_{i-1}^{-1}(z_i) - z_i) \times \frac{1}{\alpha(1-(T_{i-1}^{-1})'(z_i))+1} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } -1 \leq \alpha < 0 \end{cases}. \quad (31)$$

With this, we have all the needed terms to calculate the left terms of (27). Now we calculate the right term of (27):

$$\begin{aligned} \frac{\partial S_{N,\alpha}(y)}{\partial y} &= \frac{1}{N} \sum_{i=1}^N \frac{\partial}{\partial y} T_i \circ [\alpha T_{i-1}]^{-1}(y) \\ &= \frac{1}{N} \sum_{i=1}^N T'_i(z_i) \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)} \times \frac{1}{\frac{\partial [\alpha T_{i-1}](z_i)}{\partial z_i} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}} \\ &= \begin{cases} \frac{1}{N} \sum_{i=1}^N T'_i(z_i) \times \frac{1}{\alpha(T'_{i-1}(z_i)-1)+1} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } 0 < \alpha \leq 1 \\ \frac{1}{N} \sum_{i=1}^N T'_i(z_i) \times \frac{1}{\alpha(1-(T_{i-1}^{-1})'(z_i))+1} \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } -1 \leq \alpha < 0 \end{cases}. \end{aligned} \quad (32)$$

By plugging all the terms calculated above in (27) we get

$$\begin{aligned} \frac{\partial}{\partial \alpha} S_{N,\alpha} \circ [\alpha T_j](x) &= \begin{cases} \frac{1}{N} \sum_{i=1}^N T'_i(z_i) \times \frac{1}{\alpha(T'_{i-1}(z_i)-1)+1} \times (z_i - T_{i-1}(z_i) + T_j(x) - x) \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } 0 < \alpha \leq 1 \\ \frac{1}{N} \sum_{i=1}^N T'_i(z_i) \times \frac{1}{\alpha(1-(T_{i-1}^{-1})'(z_i))+1} \times (T_{i-1}^{-1}(z_i) - z_i + x - T_j^{-1}(x)) \Big|_{z_i=[\alpha T_{i-1}]^{-1}(y)}, & \text{for } -1 \leq \alpha < 0, \end{cases} \end{aligned} \quad (33)$$

where $y = [\alpha T_j](x)$.

The differentiability of M_N with respect to α follows from the equation above. Similarly, if we replace $S_{N,\alpha}$ with S_α , the summations can be replaced by an integral, and we can see that M is also differentiable with respect to α . Let

$$g'(T_{j-1}, T_j, S_{N,\alpha}, \alpha) = \frac{\partial g_\alpha(T_{j-1}, T_j, S_{N,\alpha})}{\partial \alpha}, \quad g'(T_{j-1}, T_j, S_\alpha, \alpha) = \frac{\partial g_\alpha(T_{j-1}, T_j, S_\alpha)}{\partial \alpha}.$$

We can argue g' is stochastically Dini-continuous when $\{T_i\} \subset \mathcal{T}_{l,u}$ (similar to the arguments in the proof of Lemma 6.11) and its derivative with respect to α is bounded. Therefore the assumptions of Corollary 6.8 (CLT) are satisfied for g' , and we have $\mathbb{E}|M'_N(\alpha) - M'(\alpha)| \lesssim \frac{1}{\sqrt{N}}$. ■

Proof of Theorem 3.11 (Convergence Rate). Using Theorem 6.12, we can obtain a rate of convergence for our estimator. First, it should be noted that the functional M is twice differentiable with respect to α since it is a composition of twice differentiable functions. As α is the unique minimizer of M , its first derivative vanishes at α , which implies that M has quadratic growth around α . Next, we need to find a function $\phi_N(\delta)$ such that

$$\mathbb{E} \sup_{|\alpha - \alpha| \leq \delta} \sqrt{N} |(M_N - M)(\alpha) - (M_N - M)(\alpha)| \leq \phi_N(\delta). \quad (34)$$

By mean value theorem, we can write:

$$(M_N - M)(\alpha) = (M_N - M)(\alpha) + (M_N - M)'(c) \times (\alpha - \alpha), \quad (35)$$

for some c between α and α . By Lemma 6.13 we can see that

$$\mathbb{E}|M'_N(c) - M'(c)| \lesssim \frac{1}{\sqrt{N}}.$$

By plugging the inequality into the expression (34) we obtain

$$\mathbb{E}\sqrt{N} |(M_N - M)(\alpha) - (M_N - M)(\alpha)| \leq |\alpha - \alpha|.$$

And, we conclude $\phi_N(\delta) = \delta$ and the rate of convergence for $\hat{\alpha}_N$ is $N^{-\frac{1}{2}}$. Using Lemma 6.2 we can see

$$\|S_{N,\hat{\alpha}_N} - \mathbf{S}\|_2 \leq \|S_{N,\hat{\alpha}_N} - S_{N,\alpha}\|_2 + \|S_{N,\alpha} - \mathbf{S}\|_2 \lesssim N^{-\frac{b}{2}} + N^{-\frac{1}{2}} \lesssim N^{-\frac{b}{2}},$$

and since $\{T_i\} \subset \mathcal{T}_{l,u}$, $b = 1$ according to Lemma 6.2. ■

6.2. Generalization of Iterated System (5)

The definition of the iterated system (4) is based on the contraction of maps around the identity map. It extends system (5) by introducing the map S . However, we could alternatively generalize (5) by introducing S not at the level of the iteration itself, but rather at the level of the contraction itself: contracting around an arbitrary map S , instead of the identity. Specifically, define the α -contraction of a map T around an arbitrary map S as follows:

$$\alpha[T, S](x) := \begin{cases} S(x) + \alpha(T(x) - S(x)) & 0 < \alpha \leq 1 \\ S(x) & \alpha = 0 \\ S(x) + \alpha(S(x) - T^{-1}(x)) & -1 \leq \alpha < 0. \end{cases} \quad (36)$$

With this definition, the original contraction operation (2) now corresponds to $\alpha[T, \text{id}]$, for $\text{id}(x) = x$ the identity map. Definition (36) directly leads to the following extension of system (5)

$$T_i = T_{\epsilon_i} \circ \alpha[T_{i-1}, S], \quad (37)$$

where $\{T_{\epsilon_i}\}_{i=1}^N$ is again a collection of i.i.d random optimal maps satisfying $\mathbb{E}\{T_{\epsilon_i}(x)\} = x$ almost everywhere on Ω . Compared to system (4),

$$T_i = T_{\epsilon_i} \circ S \circ \alpha[T_{i-1}, \text{id}].$$

this system interjects S at the level of the contraction and not at the level of the random perturbation (note that for identifiability reasons it does not make sense to do both). Of course, either is more general than system (5)

$$T_i = T_{\epsilon_i} \circ \alpha[T_{i-1}, \text{id}].$$

Remark 6.14. Suppose we use the contraction definition (36), and define the iteration (37). Then, the quantile model (UQ) with $S = F_\mu^{-1}$ (i.e., where we contract around the quantile function of a measure μ) is equivalent to the generalized quantile model (GQ) with $S = \text{id}$; that is, they produce the same stationary time series. To demonstrate this equivalence, consider the model (GQ) with $S = \text{id}$. We then have:

$$\mathbb{E}(F_{\mu_i}^{-1} \circ F_\mu(x) | F_{\mu_{i-1}}^{-1} \circ F_\mu) = \mathbb{E}(F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1} \circ F_\mu) \circ F_\mu(x) = x + \alpha(F_{\mu_{i-1}}^{-1}(F_\mu(x)) - x).$$

Thus,

$$\mathbb{E}(F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1} \circ F_{\mu}^{-1}) = F_{\mu}^{-1}(x) + \alpha(F_{\mu_{i-1}}^{-1}(x) - F_{\mu}^{-1}(x)),$$

which is equal to the conditional expectation of $\mathbb{E}(F_{\mu_i}^{-1} | F_{\mu_{i-1}}^{-1})$ when we use model (36) for $F_{\mu_i}^{-1}$ and contract around $S = F_{\mu}^{-1}$.

Remark 6.15. Note that $\alpha[T, S] = T$ when $\alpha = 1$, and $\alpha[T, S] = T^{-1}$ when $\alpha = -1$. Therefore in either of these cases, the time series T_i does not provide any information about S and it would be impossible to estimate the map S . Therefore we assume $-1 < \alpha < 1$. This is in contrast with system (4), where consistent estimation is possible for all values of $0 \leq \alpha \leq 1$,

If a stationary solution to system (37) exists, then

$$\mathbb{E}[T_i] = \mathbb{E}[T_{i+1}] = \mathbb{E}[E[T_{i+1} | T_i]] = \mathbb{E}[\alpha[T_i, S]],$$

and therefore $\mathbb{E}[T_i] = S$, when $-1 < \alpha < 1$.

We define the estimators $(\hat{\alpha}_N, S_N)$ of (α, S) as follows:

$$\hat{\alpha}_N := \arg \min_{\alpha} M_N(\alpha),$$

where

$$\begin{aligned} M_N(\alpha) &:= \frac{1}{N} \sum_{i=1}^N g(T_{i-1}, T_i, S_N) \\ g_{\alpha}(T_{i-1}, T_i, S) &:= \|\alpha[T_{i-1}, S] - T_i\|_2^2 \\ S_N &:= \frac{1}{N} \sum_{j=1}^N T_j. \end{aligned}$$

It is worth noting that unlike in system (4), where the estimation of the map S depends on the estimator of α , in this system, the estimator of the map S is simply the average of the maps T_i . Consequently, the statistical analysis of the estimators is somewhat easier in this case. Similar procedures to those used for model (4) can be used to demonstrate the existence of a unique stationary solution, the consistency of the estimator, and obtain the rate of convergence.

Assuming that system (37) satisfies the moment contracting condition 3.3, a unique stationary solution for this system exists, and $\mathbb{E}[T_i] = S$, as in the previous case. We can then use Lemma 6.1 to obtain the central limit theorem (CLT) for S_N and show that S_N converges in probability to the true S .

It is worth noting that the Lipschitz continuity property of the new function g with respect to α can be shown using the fact that $\|\alpha_1[T, S] - \alpha_2[T, S]\|_2 \lesssim |\alpha_1 - \alpha_2|$. Using this property and following a similar proof technique as in Theorem 3.11, we can argue that the rate of convergence is $N^{-1/2}$.

Remark 6.16. Once again we can use the system (37) to construct a Markov chain model for a dependent sequence of probability distributions $\mu_i \in \mathcal{W}_2(\Omega)$ by either interpreting the maps as consecutive optimal maps between a time series of probability distributions or directly using the maps to model the quantile functions. While using system (4), the increment interpretation using $\alpha = 0$ is equivalent to quantile interpretation using $\alpha = 1$, a similar straightforward relationship does not appear to exist when using system (37).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available at [https://www.ncei.noaa.gov/cdo-web/search?datasetid\\$=GHNCND](https://www.ncei.noaa.gov/cdo-web/search?datasetid$=GHNCND). The data that support the findings of this study are available in National Centers for Environmental Information at <https://www.ncei.noaa.gov>. These data were derived from the following resources available in the public domain: – National Centers for Environmental Information, <https://www.ncei.noaa.gov>.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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